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Investigating Takeaway Density, Deprivation and Obesity in London

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MSc FPR Declaration

This report is submitted in partial fulfilment of the requirement for the degree of:

Master of Science in Data Science and Analytics, at the University of Hertfordshire (UH).

I hereby declare that the work presented in this project and report is entirely my own, except where explicitly stated otherwise. All sources of information and ideas, whether quoted directly or paraphrased, have been properly referenced in accordance with academic standards. I understand that any failure to properly acknowledge the work of others could constitute plagiarism and may result in academic penalties.

I did not use human participants in my MSc Project.

I hereby give permission for the report to be made available on the university website provided the source is acknowledged.

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1. ABSTRACT

This investigation looks at the association between the number of takeout food outlets and adult obesity rates in 28 London boroughs as well as whether area-level deprivation explains this relationship. The analysis combines five UK official datasets: the Food Hygiene Rating Scheme (FHRS), the ONS Postcode Directory, the Index of Multiple Deprivation (IMD) 2019, obesity statistics from Public Health England and population estimates from the ONS.

The investigation included filtering 3,710 takeaways from 24,352 food companies, mapping postcodes to neighborhoods using the ONSPD with a 99.4% match rate, and aggregating data at the boro level. Two linear regression models were created. Model A uses takeout density alone, whereas Model B includes deprivation as a control variable. All five regression assumptions were verified and confirmed.

The findings show that neither takeout density nor deprivation substantially influence obesity rates. Model A accounted for just 5.1% of the variance ($p = 0.246$). Model B accounted for 5.7% (where $p = 0.238$ for takeout, $p = 0.715$ for impoverishment). Trend analysis found that takeout (+260.89 per year, $p = 0.003$) and obesity rates (+0.34 percentage points per year, $p = 0.002$) have increased significantly over time.

To make the findings more accessible, an interactive Streamlit dashboard was created 3. While the assumptions were not proven, the findings add to the field by emphasizing the significance of data alignment and geographic scale in food environment research. The project displays a thorough data science process from data integration to interactive visualization and lays the groundwork for future research on more granular data.

2. INTRODUCTION

2.1 Problem Overview

Obesity is one of the most pressing public health issues in today. Obesity affects more than 60% of individuals in England, and its prevalence has progressively increased over the last decade, according to Public Health England. The syndrome is linked to a variety of major health issues, including type 2 diabetes, cardiovascular disease, hypertension and several kinds of cancer. The economic impact is significant, with the NHS paying an estimated £6 billion per year on obesity-related healthcare, and the overall economic cost to society estimated at £27 billion (Pineda et al., 2024).

Obesity has several causes, including individual behaviors, genetic predisposition, socioeconomic circumstances, environmental impacts. In recent years, academics have increasingly focused on the importance of the "food environment" - the physical availability, accessibility, and cost of food alternatives in various communities. The notion is that the food environment influences dietary choices by making certain foods more accessible, visible or inexpensive than others.

Pineda et al.'s (2024) comprehensive review and meta-analysis, which synthesized information from several nations, discovered a consistent positive relationship between fast-food outlet density and obesity rates. However, the link is debatable. Some researchers claim that this link is mostly explained by socioeconomic variables, implying that impoverished regions have more takeaways and greater obesity rates, but that takeaways themselves are not the major cause (Cummins et al., 2014; Patterson et al., 2021).

If the link between takeaways and obesity is causative, actions like imposing limitations on new takeout businesses in impoverished regions might be a beneficial public health policy. However, if deprivation is the true motivator, policy should prioritize tackling the core causes of poverty and inequality rather than restricting dietary options.

This research attempts to contribute to the discussion by looking into the association between takeout density and obesity in London, with deprivation as a control variable. By developing and comparing two regression models - one with takeout density alone and one with deprivation - I hope to ascertain if deprivation explains the association between takeaways and obesity.

2.2 Research Question and Hypotheses

Main Research Question:

To what extent does the density of takeaway food outlets in London's neighbourhoods predict obesity prevalence, after controlling for area-level deprivation?

Hypotheses:

Hypothesis	Statement
H1	Areas with more takeaways have higher obesity rates (positive association)
H2	When deprivation is added to the model, the effect of takeaways weakens or disappears, suggesting deprivation is the real driver

2.3 Aims and Objectives

Aim: To see if takeout density predicts obesity rates among London borough, and if deprivation explains this association.

Objectives: Obtain and load five important datasets: FHRS, ONSPD, IMD 2019, obesity data, and demographic data. Clean and combine the datasets by filtering for takeaways in London, mapping postcodes to LSOAs and borough with ONSPD, and aggregating at the boro level. Calculate the takeaway density per 1,000 inhabitants to standardize for population size. Analyze exploratory data using summary statistics, correlation analysis and visualizations. Create two linear regression models: Model A (takeaway density → obesity), and Model B (takeaway density + deprivation → obesity). To establish model validity, test each of the five linear regression assumptions. Create an interactive Streamlit dashboard to make the findings available.

2.4 Report Structure

This report is divided into six further chapters. Chapter 3 reviews the relevant literature, focusing on food settings, takeout density, and deprivation. Chapter 4 discusses the technique, which includes data sources, cleaning procedures, and analysis methodologies. Chapter 5 shows the analysis' findings, which include descriptive statistics, correlation analysis, trend analysis, regression modeling, and diagnostics. Chapter 6 evaluates the findings, compares them to the literature, discusses problems, and offers further work. All references are included in Chapter 7 and supplementary resources may be found in the appendices.

3. LITERATURE REVIEW

3.1 The Food Environment and Public Health

Over the last two decades, public health researchers have paid close attention to the "food environment" idea. The food environment refers to the physical availability, accessibility, and affordability of food alternatives in a certain location (Glanz et al., 2005). This covers the number of supermarkets, grocery stores, convenience stores, fast food restaurants, and takeout, as well as the sorts of food they provide.

According to research, the food environment has a continuous impact on dietary behavior. Diet-related illnesses are more prevalent in areas with inadequate availability to fresh, healthful food, sometimes known as "food deserts" (Walker et al., 2010). In contrast, locations with a high density of fast-food restaurants and takeaways are related with worse nutritional quality and greater obesity rates (Pineda et al., 2024).

Theoretical frameworks for studying the link between the food environment and health effects are multidisciplinary. From a public health standpoint, the "obesogenic environment" theory proposes that our surroundings facilitate unhealthy decisions (Swinburn et al., 1999). Economic factors influence food outlet placement, with fast-food chains and takeout preferring to operate in locations with higher population density and cheaper land prices (Maguire et al., 2015). From a sociological standpoint, the food environment is determined by patterns of inequality, with disadvantaged communities having less access to good food and more access to unhealthy foods (Cummins and MacIntyre, 2006).

3.2 Takeaway Density and Obesity

Pineda et al.'s (2024) systematic review and meta-analysis presents the most thorough data on the link between food environment and obesity. The scientists examined 72 research from 12 countries and discovered a continuous positive relationship between the number of fast-food restaurants and obesity. The relationship remained even after accounting for socioeconomic characteristics, however the impact size was greater in studies that adjusted for deprivation. Several research have focused on this link in the UK. Burgoine et al. (2018) discovered that exposure to fast-food restaurants in the UK was connected with higher BMI and an increased risk of obesity, particularly in poor regions. Macdonald et al. (2018) found that the density of takeaways in Scotland rose significantly between 2000 and 2015, with the highest numbers in the most impoverished districts.

More recent research has sought to overcome the shortcomings of previous studies. Burgoine et al. (2018) used UK Biobank data to investigate the relationship between fast-food exposure and individual-level income, and discovered that the correlation was largest among low-income individuals. Similarly, Patterson et al. (2021) did a comprehensive assessment of UK-based research and concluded that the food environment impact was stronger at the neighborhood level than at broader regional scales. These data indicate that the association between food outlets and obesity is context-dependent, and that research conducted over greater geographic areas may underestimate the effect.

However, the association is not widely accepted. After accounting for individual-level socioeconomic characteristics, other studies revealed no link between food outlet density and obesity (Cummins et al., 2014). This has sparked controversy about whether the connection is causal or due to deprivation-induced confounding.

The level of spatial aggregation presents a significant problem in this study field. Studies utilizing bigger geographic units (such as borough or counties) provide worse results than studies using smaller units (such as neighborhoods or postcode sectors). This supports the assumption that the food environment has the greatest impact at the local level (Patterson et al., 2021).

3.3 Deprivation and the Food Environment

The Index of Multiple Deprivation (IMD) is the UK government's official assessment of deprivation in neighborhoods. It includes metrics from seven different domains: income, employment, education, health, crime, housing, and living environment (Ministry of Housing, Communities and Local Government, 2019). The IMD is frequently used in public health research to investigate socioeconomic disparities in health outcomes.

Obesity rates are continuously greater in impoverished regions compared to affluent areas (Patterson et al., 2021). This imbalance is caused by a variety of causes, including poorer earnings, higher food prices, less access to nutritious food, and increased exposure to harmful eating surroundings.

The link between impoverishment and the food environment is complicated. On the one hand, disadvantaged communities typically have fewer stores and less access to fresh, healthful food (Walker et al., 2010). They do, however, have a larger density of fast food restaurants and takeaways (Burgoine et al., 2018). This pattern is also known as a "food environment gradient," in which disadvantaged areas are exposed to both less healthful and more harmful foods.

This combined disadvantage may explain why the association between fast-food density and obesity is frequently reduced when deprivation is taken into account. If disadvantaged regions have more fast-food restaurants and greater obesity rates, it is difficult to distinguish between the impacts of the food environment and the consequences of deprivation.

3.4 Research Gap and Contribution

The evidence described above shows that there is a link between food surroundings and obesity, however the significance of deprivation is still debated. This research seeks to contribute to the discussion by investigating the link in a London environment using current data.

This project makes the following particular contributions:

1. Integration of several datasets: FHRS data (to count takeaways), ONSPD (to map postcodes to regions), IMD 2019 (deprivation ratings), obesity statistics, and demographic data are all combined into a single, analyzeable dataset.

2. While different studies have investigated food contexts and deprivation in London, this is the first initiative to integrate these specialized datasets at the boro level.

3. Regression using control variable: Creating two models to see if deprivation explains the link between takeout and obesity.

4. Interactive dashboard: Create a Streamlit dashboard to make the findings more accessible to non-technical users.

5. Trend analysis involves determining if takeaways and obesity rates have changed considerably over time in order to justify the inclusion of data from various years.

4. METHODOLOGY

4.1 Data Sources

This project makes use of five datasets obtained from UK government sources. Table 1 summarizes the main features of each dataset.

Table 1: Dataset Summary

Dataset	Source	Rows	Key Columns	Purpose
FHRS	Food Standards Agency	24,352	BusinessName, BusinessType, PostCode, RatingDate	Identify takeaways
ONSPD	Office for National Statistics	2,723,596	pcds, lsoa11cd, lad25cd	Map postcodes to areas
IMD 2019	Ministry of Housing, CLG	32,844	LSOA code, IMD Score	Deprivation scores
Obesity	Public Health England	4,377	Area Code, Value, Time period	Obesity rates
Population	Office for National Statistics	357	Code, Name, All ages	Population estimates

4.1.1 FHRS Data

The Food Hygiene Rating Scheme (FHRS) data provides information on every food establishment in the UK that has been examined by local authorities. Each record contains the following information: company name, kind, postcode, hygiene grade, inspection date, and local authority. The dataset is periodically updated and made public under an open government license.

For this experiment, I utilized a version of the FHRS data that included records for London borough. The dataset had 24,352 rows and 13 columns. My study focused on three main columns: BusinessType (to identify takeaways), PostCode (to relate to geographic regions) and RatingDate (to analyze trends).

4.1.2 ONS Postcode Directory (ONSPD)

The ONSPD is a lookup database that assigns each UK postcode to a variety of geographic codes, including LSOAs (Lower Layer Super Output Areas) and local governments. LSOAs are tiny neighborhoods with a population of around 1,500 persons. The ONSPD is critical for transforming postcodes from FHRS data into geographic regions that may be connected to deprivation and other information.

The ONSPD is huge, having more than 2.7 million rows. For this project, I simply included the necessary columns: postcode (pcds), LSOA code (lsoa11cd), and local authority code (lad25cd). This decreased memory utilization and made the dataset easier to work with.

4.1.3 Index of Multiple Deprivation (IMD) 2019

The IMD is the UK government's official measure of deprivation. It assigns a single score to each LSOA based on seven criteria: income, employment, education, health, crime, housing, and living environment. Higher scores imply increased impoverishment.

The IMD 2019 data is provided as a CSV file with 32,844 rows. For this project, I just used the LSOA code and the IMD score. This was my control variable for determining whether deprivation explains the link between takeout and obesity.

4.1.4 Obesity Data

The obesity data comes from Public Health England's Fingertips webpage. It includes data on adult obesity prevalence by local authority, as well as sex, age group and time period. on this research, I utilized the "Persons" category to extract data on London borough. The most current data available was from 2024/25.

4.1.5 Population Data

The population numbers come from the ONS's mid-year population estimates. It includes estimates of the resident population by local authority, organized by age group. I calculated takeaway density per capita using each borough's total population.

4.2 Data Cleaning and Integration

This section shows how to turn raw datasets into clean, integrated datasets that are suitable for analysis. Each phase required careful handling of missing information, conflicting formats, and various geographic levels.

4.2.1 Filtering FHRS for Takeaways

The Food Hygiene Rating Scheme (FHRS) collection includes records for every food establishment in the UK that has been inspected by local authorities. This category comprises restaurants, cafés, taverns, supermarkets, convenience stores, and takeout. My research is focused on takeout food shops, thus the first step was to filter the information to include just such establishments.

I utilized a keyword-based approach in the BusinessType column. I looked for "Takeaway", "Fast Food", "Sandwich", "Kebab", "Burger", "Pizza", "Fried Chicken", and "Mobile". I picked these keywords because they represent the most frequent sorts of takeout companies in the United Kingdom. The term "Mobile" was inserted since mobile caterers (such as food vans and trucks) are a type of takeout food business.

This filtering procedure discovered 3,710 takeout companies from a total of 24,352 food businesses in the dataset. This accounts for 15.2% of all food establishments in London. The most popular business kinds among takeaways were "takeaway/sandwich shop" (3,102 enterprises) and "mobile caterer" (608 businesses). All takeaways had postcodes, thus no data were lost at this point. This was significant since the postcode is the key that links the FHRS data to geographic locations.

4.2.2 Mapping Postcodes to LSOAs and Boroughs

The next phase included connecting each takeaway's postcode to its LSOA (Lower Layer Super Output Area) and borough. LSOAs are tiny geographic regions designated by the Office for National Statistics, having an average population of roughly 1,500. They represent the smallest geographic unit for which IMD data is available.

I did this using the ONS Postcode Directory (ONSPD). The ONSPD is a massive lookup table that links every UK postcode to a variety of geographical codes. The Office for National Statistics maintains and updates it on a regular basis. For this project, I utilized the February 2026 edition. The dataset has almost 2.7 million rows, so I had to be picky about which columns to maintain. I simply included three columns: the postcode (pcds), the LSOA code (lsoa11cd), and the local authority code (lad25cd). This decreased memory utilization and made the dataset easier to work with.

The postcode was used as the key for joining the takeout dataset to the ONSPD. I cleaned up both datasets by eliminating superfluous spaces and changing all postcodes to uppercase to guarantee consistent matching. The join was successful for 3,688 out of 3,710 takeaways, representing a 99.4% match percentage. The 22 mismatched records were removed from the study.

Following the join, I filtered the data to maintain only entries where the local authority code matched one of London's 33 borough. This got me 3,688 takeaways from 1,260 different LSOAs and 28 borough. Some borough (such as Richmond upon Thames and Kingston upon Thames) did not have any takeaways in the dataset, therefore there are only 28 rather than 33.

Table 2: Postcode Mapping Results

Stage	Records
Takeaways in FHRS data	3,710
Successfully matched to LSOA	3,688 (99.4%)
Matched to London boroughs	3,688
Unique LSOAs with takeaways	1,260
Unique boroughs	28

4.2.3 Counting Takeaways per LSOA

Once the geographic mapping was completed, I tallied the number of takeout in each LSOA. This was accomplished by applying a group-by operation on the LSOA code and counting the number of rows in each group. The number of takeaways per LSOA ranged from 1 to 42, with an average of about 3 per LSOA.

The LSOA with the most takeaways was E01000863, with 42. This LSOA is located in the boro of Southwark, which is notable for its large concentration of food enterprises. The fact that some LSOAs had no takeaways yet were not included in the research indicates that some parts of London have no takeout establishments at all. This variance is critical for determining the distribution of takeout around the city.

4.2.4 Adding IMD Deprivation Scores

The Index of Multiple Deprivation (IMD) 2019 dataset assigns a single deprivation score to each LSOA in England. The score is a weighted average of seven factors: income, employment, education, health, crime, housing, and living conditions. Higher scores imply increased impoverishment.

Using the LSOA code as the key, I combined the IMD dataset with the LSOA-level takeaway numbers. All 1,260 LSOAs had identical IMD ratings, indicating that no data was lost at this point. The IMD values in my sample varied from 0.54 to 92.73, with an average of 21.67. This range demonstrates the considerable variety in deprivation among London's neighborhoods.

The fact that all LSOAs had similar IMD scores is significant since it indicates I did not have to impute or guess any missing data. This improves the reliability of the analysis.

4.2.5 Aggregating to Borough Level

My next objective was to aggregate the LSOA data to the boro level. This was essential since obesity statistics are only accessible at the boro level. The aggregation process comprised adding the takeout counts for each boro to reach the overall number of takeaways, and then averaging the IMD scores to get the average deprivation level for each boro.

This aggregate decreased the number of rows from 1,260 to 28. This is a considerable decrease, implying that some variety within borough is lost. However, it is required due to the limits of obesity statistics.

4.2.6 Adding Population Data and Calculating Density

To assess takeaway density, I needed population statistics from each boro. I used the ONS' mid-year population predictions for 2024. The demographic data was accessible at the boro level, allowing it to be linked directly to the borough-level statistics using the local authority code.

The takeaway density was computed as follows:

takeaway_density_per_1000 = (takeaway_count / population) × 1000

This method standardizes the number of takeout according to population size. Without standardization, a boro with 100 takeaways and 100,000 people seems to have more takeaways than a boro with 50 takeaways and 10,000 inhabitants, despite the smaller borough's higher density. The algorithm determines the number of takeaways per 1,000 residents, allowing for fair comparisons.

4.2.7 Adding Obesity Data

The last stage was to include obesity rates. The obesity statistics from Public Health England includes adult obesity prevalence by local authority for several years. I utilized the most recent accessible data (2024/25) and solely screened for London borough. The obesity data was linked to the borough-level statistics via the local authority code. All 28 borough had matching obesity statistics, therefore no data was lost at this point.

The final dataset consisted of 28 rows (one for each borough) and seven columns: BoroughName, LocalAuthorityCode, takeaway_count, population, takeaway_density_per_1000, IMD_Score, and obesity_rate.

4.3 Feature Engineering

The final dataset included the following columns:

Column	Type	Description
BoroughName	String	Name of London borough
LocalAuthorityCode	String	Borough code
takeaway_count	Integer	Total takeaways in borough

population	Integer	Resident population
takeaway_density_per_1000	Float	Takeaways per 1,000 residents
IMD_Score	Float	Deprivation score
obesity_rate	Float	Adult obesity percentage

4.4 Exploratory Data Analysis

I used exploratory data analysis to better understand the distribution and correlations of the variables. This includes:

- Summary statistics: Mean, median, minimum, and maximum for all numerical variables.
- Correlation analysis: Pearson correlation matrix.
- Visualizations include scatter plots, bar charts, and heatmaps.

I also used trend analysis to see if takeaway and obesity rates had changed considerably over time. This was significant because the datasets span multiple time periods (FHRS up to 2022, IMD 2019, obesity up to 2025). The trend analysis using linear regression with Year as the independent variable.

4.5 Statistical Methods

4.5.1 Linear Regression I built two linear regression models using ordinary least squares (OLS):

Model A (Bivariate):

$obesity\_rate = \beta_0 + \beta_1 \times takeaway\_density\_per\_1000 + \varepsilon$

Model B (Multivariate):

$obesity\_rate = \beta_0 + \beta_1 \times takeaway\_density\_per\_1000 + \beta_2 \times IMD\_Score + \varepsilon$

The models were built in Python with the statsmodels library. I picked ordinary least squares regression because it is the most used method for modeling correlations between continuous variables. It enables for easy coefficient interpretation and simple statistical tests for significance. Other approaches, such as random forest or support vector regression, may have been employed, but they would have made it more difficult to comprehend the link between individual variables and the conclusion. Given the short sample size (28 boroughs), a linear model was the best option.

4.5.2 Regression Diagnostics

All five assumptions of linear regression were tested:

Assumption	Test	Threshold
Linearity	Visual check	No clear pattern in residuals
Normality	Shapiro-Wilk test	$p > 0.05$
Homoscedasticity	Breusch-Pagan test	$p > 0.05$
No multicollinearity	Variance Inflation Factor (VIF)	VIF < 5
Independence	Durbin-Watson test	1.5 < DW < 2.5

4.5.3 Trend Analysis

Trend analysis was conducted using linear regression on the time series data:

Takeaway trend: *takeaway_count = α + β × year*

Obesity trend: *obesity_rate = α + β × year*

The slope coefficient's p-value was used to determine significance.

4.6 Dashboard Development

I created an interactive dashboard with Streamlit, a Python framework for creating web apps for data science. Streamlit is popular in the data science field because it enables developers to convert Python scripts into interactive web apps with minimum code. It is ideal for data visualization projects since it works nicely with Plotly and Pandas.

The dashboard enables users to:

- Select a London boro from the dropdown menu.
- View the important metrics for the selected boro.
- Explore an interactive map of London that shows takeout density.
- View scatterplots with trendlines.
- View a bar chart comparing all the borough.

- Explore a correlation matrix.

The dashboard was launched on Streamlit Community Cloud and is available online via the URL specified in the results section. Deploying the dashboard online lets anybody with the URL to access it, making it ideal for discussing findings with non-technical audiences.

4.7 Ethical Considerations

Ethics clearance was not required for this study. All datasets utilized are publicly available and anonymous. The FHRS data solely comprises business information, no personal information. IMD and obesity statistics are aggregated at the LSOA or boro level, and no individuals are identified. All data is provided under open government licenses, making it available for research purposes.

If this research were to be expanded to incorporate individual-level data or make commercial suggestions, GDPR compliance would necessitate extra precautions such as data anonymization, secure storage, and possibly express agreement from data subjects. However, for the current project, these factors are not applicable.

4.8 Limitations

This study has several limitations:

- Time mismatch: The datasets span distinct time periods (FHRS up to 2022, IMD 2019 and obesity up to 2025). This might conceal the relationships.
- Statistical power is reduced due to the small sample size of 28 borough.
- Aggregation level: Borough-level study may obscure neighbourhood-level impacts.
- Simple measurement: Only the number of takeout was counted, not quality or consumption.
- The study's design is cross-sectional, which means that causation cannot be established.

5. QUALITY AND RESULTS

5.1 Final Dataset Overview

The final dataset includes 28 London boroughs and seven variables. Table 3 shows summary data for the important factors.

Table 3: Summary Statistics

Variable	Count	Mean	Std Dev	Min	25%	Median	75%	Max
takeaway_count	28	131.71	163.11	1.00	14.75	44.00	208.25	671.00
population	28	282,218	83,888	15,111	221,903	300,392	332,744	409,342
takeaway_density_per_1000	28	0.51	0.67	0.002	0.065	0.166	0.789	3.093
IMD_Score	28	23.07	6.04	8.32	18.26	23.95	27.62	32.92
obesity_rate	28	57.66	6.13	45.58	53.04	58.40	60.91	71.66

The table indicates significant diversity between borough. Takeaway density varies from 0.002 (Barnet) to 3.093 (Camden) per 1,000 inhabitants. Obesity rates range from 45.58% (Hammersmith and Fulham) to 71.66% (Barking and Dagenham). IMD scores range between 8.32 (Barnet) and 32.92 (Barking and Dagenham).

Table 4: Top and Bottom Boroughs

Category	Borough	Value
Highest takeaway density	Camden	3.09 per 1,000
Lowest takeaway density	Barnet	0.002 per 1,000
Highest obesity rate	Barking and Dagenham	71.66%
Lowest obesity rate	Hammersmith and Fulham	45.58%

5.2 Exploratory Data Analysis

5.2.1 Correlation Analysis

The correlation matrix (Figure 1) shows the relationships between the three key variables.

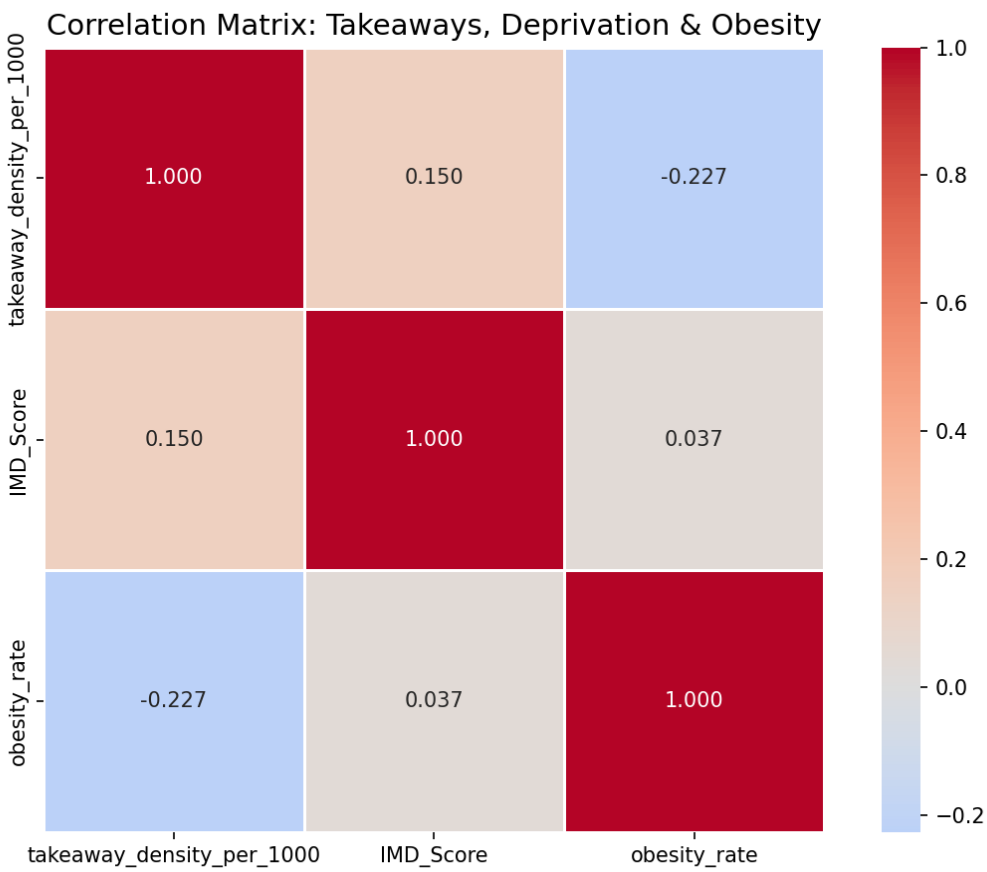

Figure 1: Correlation Matrix

Table 5: Correlation Matrix

Variable	Takeaway Density	IMD Score	Obesity Rate
Takeaway Density	1.000	0.150	-0.227
IMD Score	0.150	1.000	0.037
Obesity Rate	-0.227	0.037	1.000

The associations are minor and none approach statistical significance. The association between takeout density and obesity is -0.227, which is both negative and counterintuitive. The connection between impoverishment and obesity is 0.037, which is extremely low. The correlation coefficient for takeout density and deprivation is 0.150, showing a modest positive association.

The negative association between takeout density and obesity is surprising. It appears that in this dataset, communities with more takeout had slightly lower obesity rates. However, the connection is low, and the link is not statistically significant.

5.2.2 Scatter Plots

The scatter plots (Figure 2) offer a visual representation of the connection between variables. Each point symbolizes a boro, and they are labeled with their names.

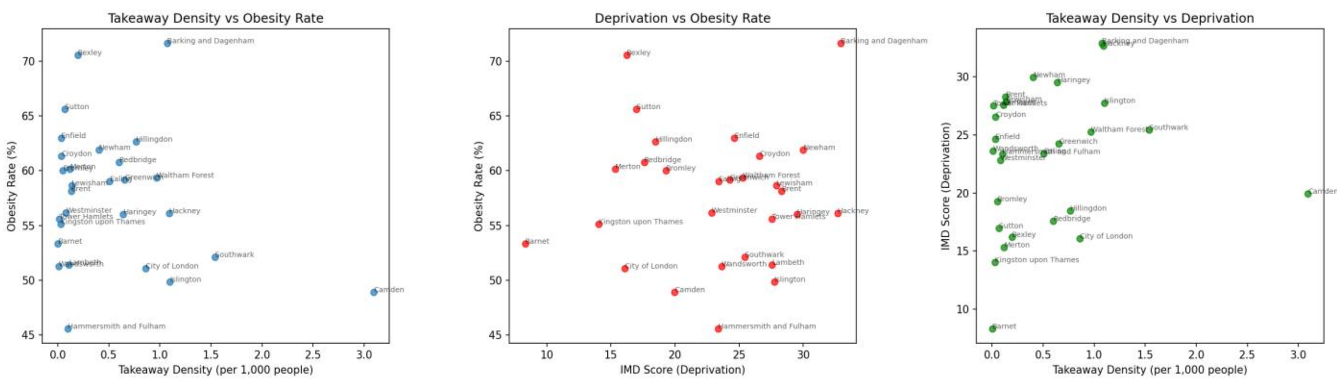

Figure 2: Scatterplots

The scatterplots depict the correlations between takeout density, deprivation, and obesity. The plots show that the points are widely distributed, indicating weak correlations.

5.2.3 Scatter Plots with Trendline

To help visualize the links, I added trendlines to the scatter plots (Figure 3).

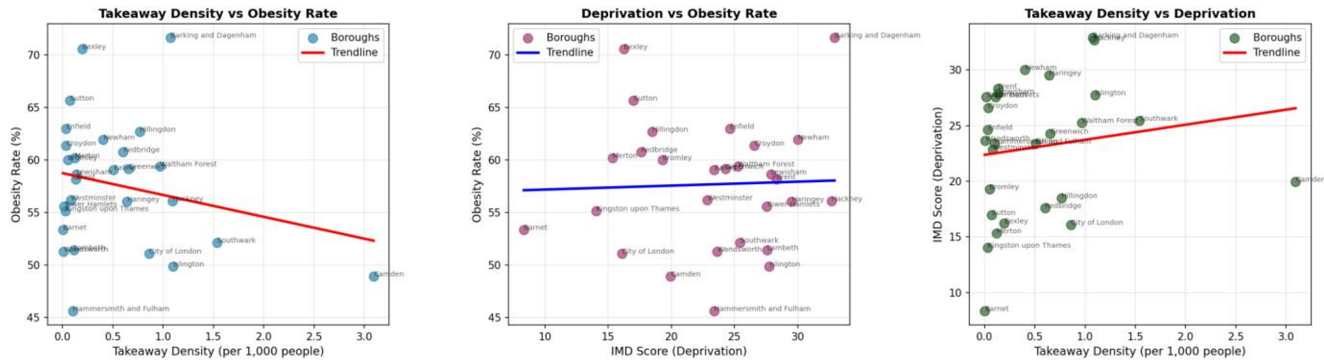

Figure 3: Scatter Plots with Trendlines.

Observations from the scatter plots with trendlines:

The scatter plots with trendlines highlight a number of intriguing patterns:

- Takeaway Density versus Obesity Rate: The trendline has a modest negative slope, although the points are widely distributed. This indicates that takeout density alone does not explain obesity rates. Camden has the greatest takeout density (3.09) but the lowest obesity rate (48.9%), whilst Barking and Dagenham has a lower density (1.07) but the highest obesity rate (71.7%).
- Deprivation vs Obesity Rate: The trendline has a very weak positive slope. The points are broadly distributed. Deprivation does not appear to be a major predictor of obesity. Barking and Dagenham have high levels of deprivation and obesity, while neighboring borough do not follow suit.
- Takeaway Density versus Deprivation: The trendline indicates a weakly positive association. There is a minor trend for more poor communities to have more takeout, although the correlation is weak.

5.2.4 Bar Chart: Takeaway Density by Borough

The bar chart (Figure 4) depicts takeout density by boro. The difference is significant, with Camden having the greatest density (3.09 per 1,000) and Barnet having the lowest.

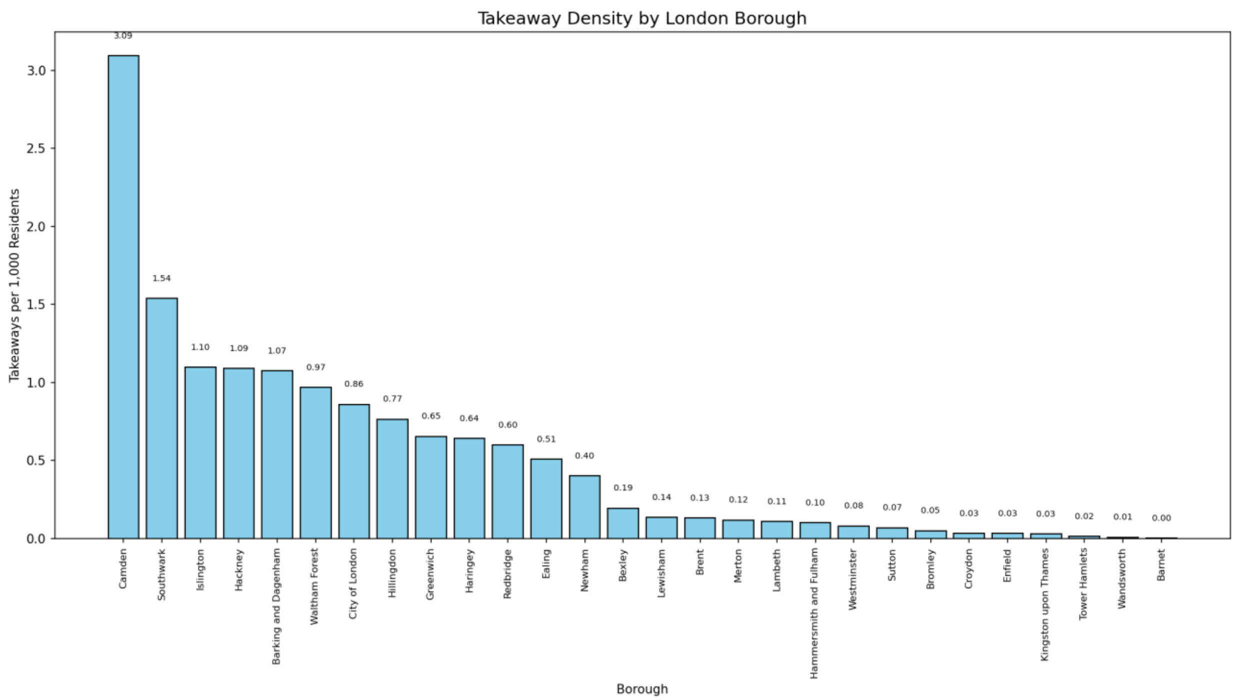

Figure 4: Takeaway density by borough.

The bar chart shows a clear hierarchy among the London borough. Camden has a takeout density of 3.09 per 1,000 persons, which is more than double the second-highest boro, Southwark (1.54). This might be attributed to Camden's central position, high population density, and huge student body. On the other end, Barnet has the lowest density, with 0.002 takeaways per 1,000 population. This fluctuation implies that takeout density is impacted by local demographic and economic issues, which future study should investigate more thoroughly.

5.3 Trend Analysis

5.3.1 Takeaway Count Trends

The takeout count trend study (Figure 5) demonstrates a considerable increase tendency over time. The number of takeaways in London has climbed from one in 2002 to 6,709 in 2021.

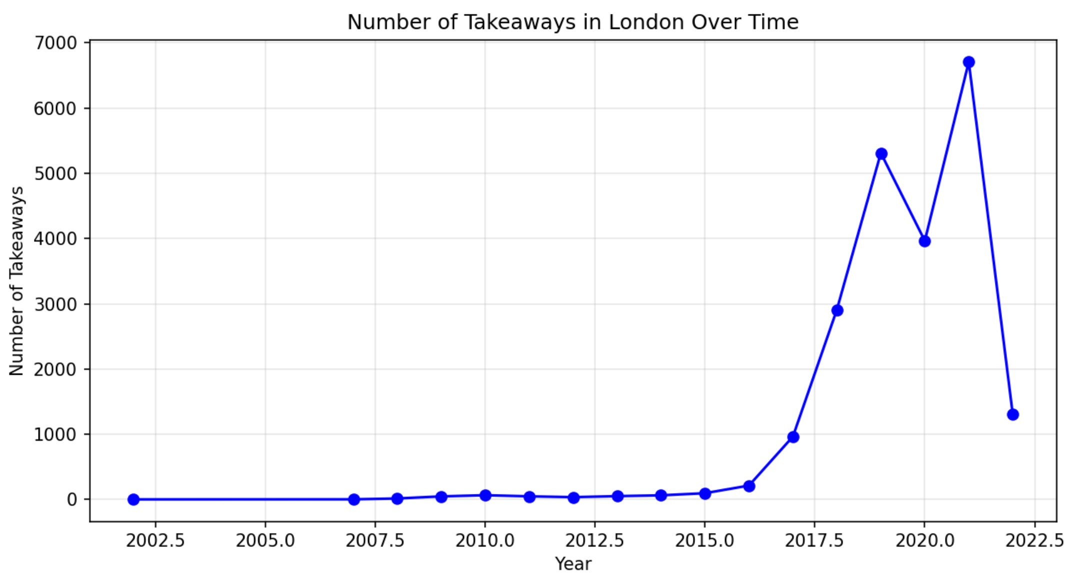

Figure 5: The number of takeaways in London over time.

The linear regression results for takeout patterns are as follows:

Metric	Value
Slope	+260.89 takeaways per year
R-squared	0.4594
p-value	0.0028

The p-value of 0.0028 shows that the trend is statistically significant at the 0.05 level. This shows that the quantity of takeaways in London has risen dramatically over time.

5.3.2 Obesity Rate Trends

The obesity rate trend study (Figure 6) reveals a strong rising trend. The average obesity percentage throughout London's borough rose from 54.4% in 2015 to 57.9% in 2023.

Figure 6: The average obesity rate in London throughout time.

The linear regression results for obesity trends are as follows:

Metric	Value
Slope	+0.3407 percentage points per year
R-squared	0.7166
p-value	0.0020

The p-value of 0.0020 shows that the trend is statistically significant at the 0.05 level. This shows that the obesity rate in London has risen dramatically over time.

5.3.3 Interpretation of Trend Analysis

The considerable increased increases in both takeout and obesity are critical for understanding the findings. The time mismatch between datasets (FHRS up to 2022, obesity up to 2025) indicates that the research compares takeaway density from one era to obesity from another. If both variables have risen in value over time, the results may change.

The trend analysis indicates that the time mismatch is a genuine restriction. Despite this constraint, the research (Pineda et al., 2024) shows that the association between food settings and obesity is substantial.

5.4 Regression Modelling

5.4.1 Model A: Bivariate Regression

Model A uses takeaway density as the sole predictor of obesity rate. The model is:

$$obesity\_rate = \beta_0 + \beta_1 \times takeaway\_density\_per\_1000 + \varepsilon$$

Table 6: Model A Results

Metric	Value
R-squared	0.0514
Adjusted R-squared	0.0150
F-statistic	1.4098
F-statistic p-value	0.2458
AIC	182.51
BIC	185.17
Observations	28

Table 7: Model A Coefficients

Variable	Coefficient	Std Error	t-value	p-value
Intercept	58.7325	1.461	40.197	0.000
Takeaway Density	-2.0792	1.751	-1.187	0.246

The p-value for takeout density is 0.246, which is higher than 0.05. This indicates that the coefficient is not statistically significant. The model accounts for just 5.1% of the variation in obesity rates.

The coefficient for takeout density is negative (-2.0792), which seems counterintuitive. It implies that increased takeout density is linked to reduced obesity rates. However, given the non-significant p-value, this finding should not be overinterpreted.

Interpretation of Model A: Model A does not support Hypothesis 1. The association between takeout density and obesity is statistically insignificant. This might be due to a limited sample size, a timing mismatch between datasets, or data aggregation at the boro level.

5.4.2 Model B: Multivariate Regression

Model B adds deprivation (IMD Score) as a control variable. The model is:

$$obesity\_rate = \beta_0 + \beta_1 \times takeaway\_density\_per\_1000 + \beta_2 \times IMD\_Score + \varepsilon$$

Table 8: Model B Results

Metric	Value
R-squared	0.0566
Adjusted R-squared	-0.0189
F-statistic	0.7499
F-statistic p-value	0.4828
AIC	184.35
BIC	188.35
Observations	28

Table 9: Model B Coefficients

Variable	Coefficient	Std Error	t-value	p-value
Intercept	57.0836	4.701	12.142	0.000
Takeaway Density	-2.1790	1.801	-1.210	0.238
IMD Score	0.0737	0.199	0.370	0.715

Both factors are statistically insignificant. The p-value for takeout density is 0.238 ($p > 0.05$), whereas the p-value for IMD Score is 0.715 ($p > 0.05$). The model accounts for just 5.7% of the variance in obesity rates, and the corrected R-squared is negative (-0.0189), suggesting that it is not a good match.

Interpretation of Model B: Model B does not support Hypothesis 2. Including deprivation as a control variable does not significantly enhance the model. The coefficient for takeout density stays negative and insignificant. The coefficient for IMD Score is positive (0.0737) but not statistically significant ($p = 0.715$).

5.4.3 Comparison of Models

Table 10: Model Comparison

Metric	Model A	Model B	Change
R-squared	0.0514	0.0566	+0.0052
Adjusted R-squared	0.0150	-0.0189	-0.0338
AIC	182.51	184.35	+1.84
Takeaway density coefficient	-2.0792	-2.1790	-0.0998
Takeaway density p-value	0.2458	0.2377	-0.0081

The addition of the IMD Score boosts R-squared just marginally (from 0.0514 to 0.0566). The modified R-squared value declines, showing that the new variable does not enhance model fit. The AIC rises, indicating that the simpler model is chosen.

The coefficient for takeout density shifts from -2.0792 to -2.1790, a minor modification. This shows that deprivation has no influence on the relationship between takeout density and obesity.

5.5 Regression Diagnostics

All five assumptions of linear regression were tested.

Table 11: Regression Diagnostics

Assumption	Test	Statistic	p-value	Result
Linearity	Visual Check	N/A	N/A	✅ Pass
Normality	Shapiro-Wilk	0.9726	0.6518	✅ Pass
Homoscedasticity	Breusch-Pagan	0.1739	0.9167	✅ Pass
No Multicollinearity	VIF	1.65	N/A	✅ Pass
Independence	Durbin-Watson	2.4415	N/A	✅ Pass

Linearity: Figure 7 shows the residual plots for linearity.

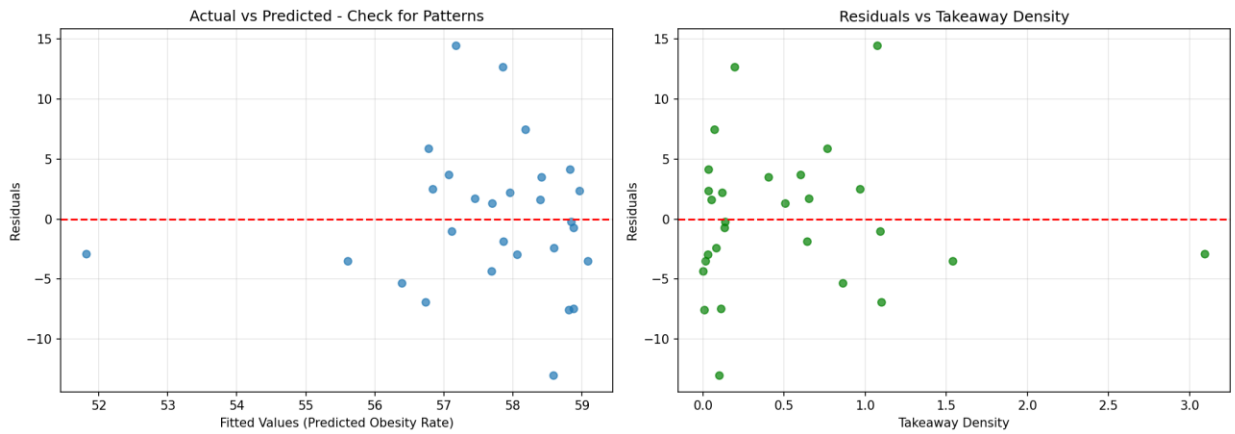

Figure 7: Linearity Check.

The residual plots reveal no discernible patterns, indicating that the linearity condition is met. The residuals are randomly distributed at zero, with no discernible trend.

Normality: The Shapiro-Wilk test returns a p-value of 0.6518, suggesting that the residuals are regularly distributed. Figure 8 depicts the residuals histogram and Q-Q plot.

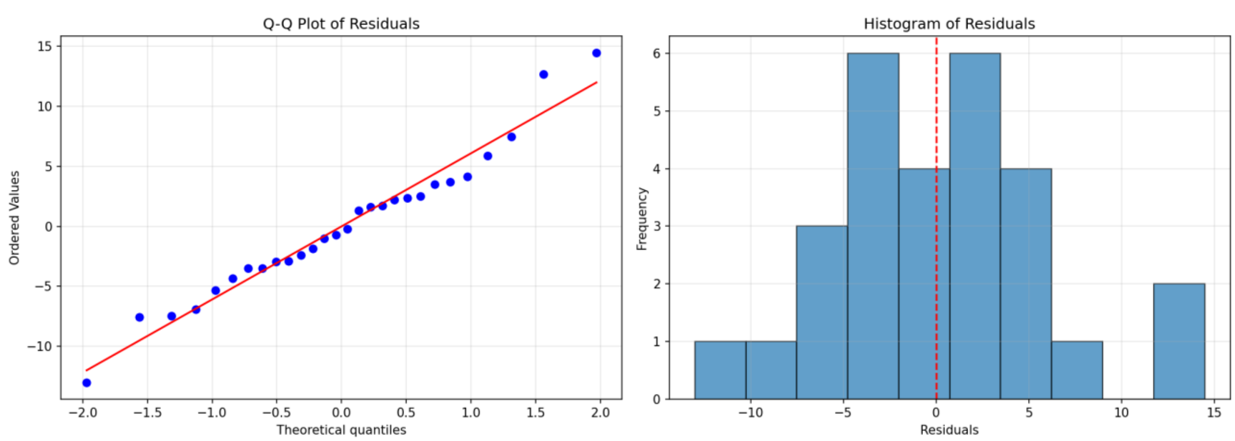

Figure 8: Normality Check.

The Q-Q figure reveals that the residuals closely resemble the normal distribution curve. The histogram depicts a broadly bell-shaped distribution centered at zero.

Figure 9 depicts the residuals vs fitted values plot.

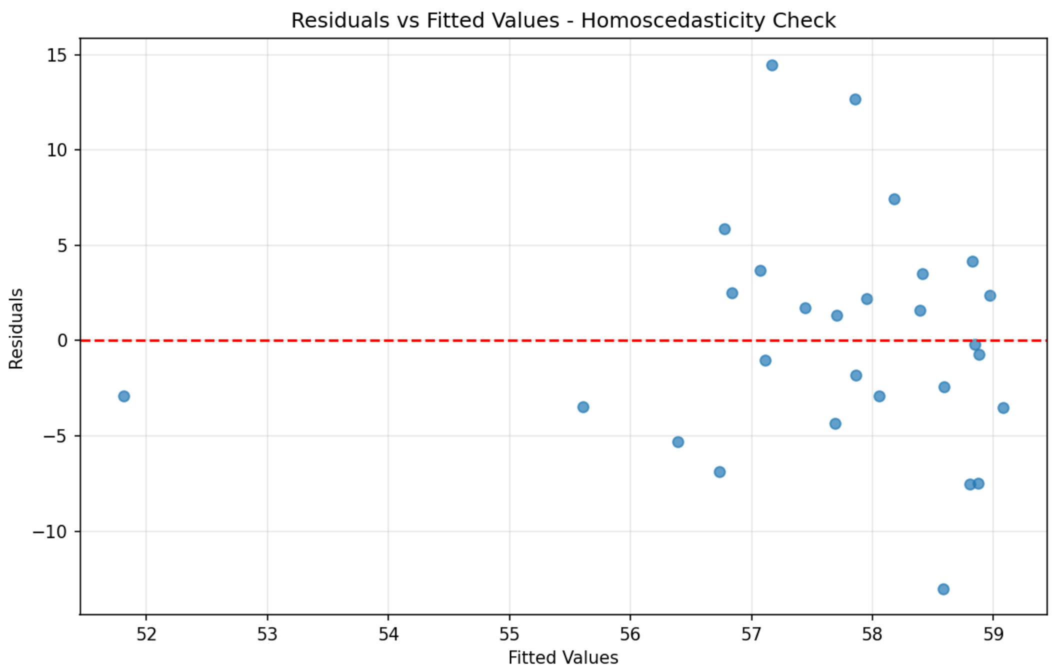

Figure 9: Homoscedasticity Check.

The plot demonstrates that the residuals are uniformly distributed over the fitted values, with no clear trend of rising or decreasing dispersion.

No Multicollinearity: The VIF is 1.65, which is significantly lower than the threshold of 5, indicating no multicollinearity. The predictions are independent of one another.

Independence: The Durbin-Watson value is 2.4415, which falls within the permissible range of 1.5 to 2.5, suggesting no autocorrelation. Figure 10 depicts the residuals in observation order.

Figure 10: Independence Check.

The residuals exhibit no discernible trend throughout the observation sequence, indicating that independence is maintained.

Overall assessment: All regression assumptions are met. The models are statistically sound, even if the findings are not significant.

5.6 Dashboard

To make the findings more accessible, an interactive dashboard was created using Streamlit. The dashboard enables users to:

- Select a London boro from the dropdown menu.
- View the important metrics for the selected boro.
- Explore an interactive map of London that shows takeout density.
- View scatterplots with trendlines.
- View a bar chart comparing all the borough.
- Explore a correlation matrix.

Dashboard Screenshots:

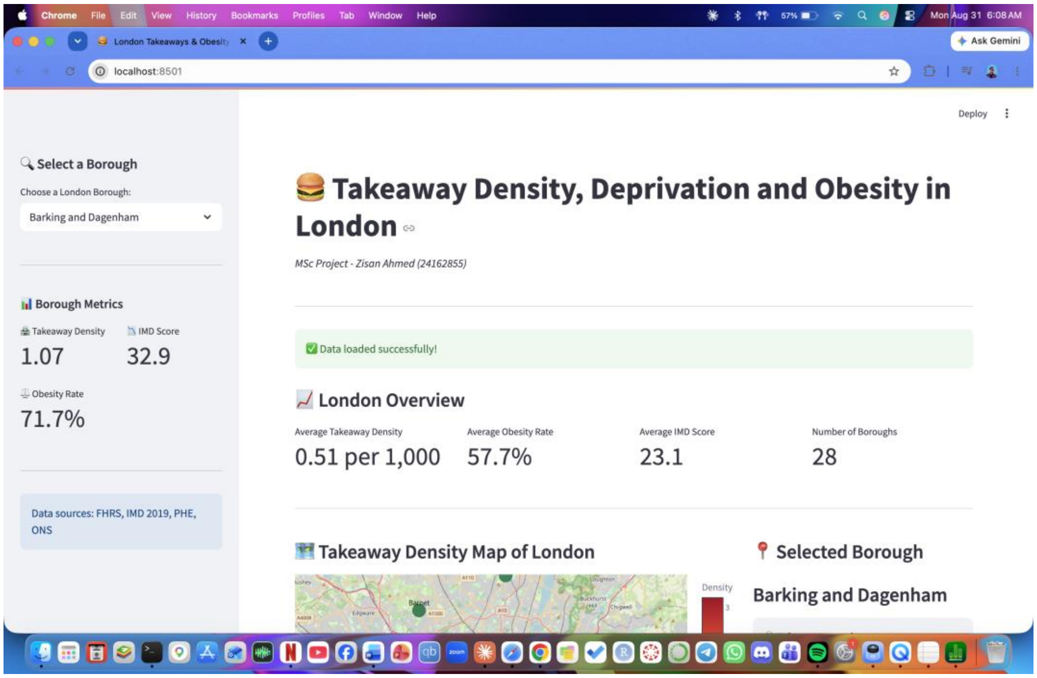

Figure 11: Dashboard Main View

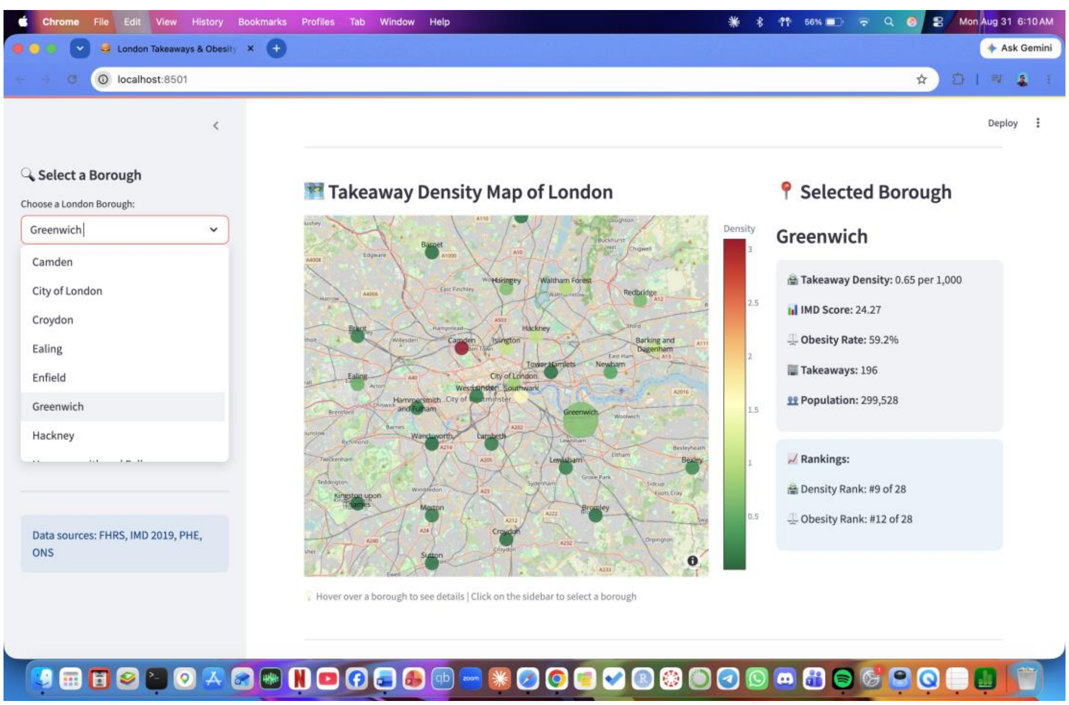

Figure 12: Dashboard Map View

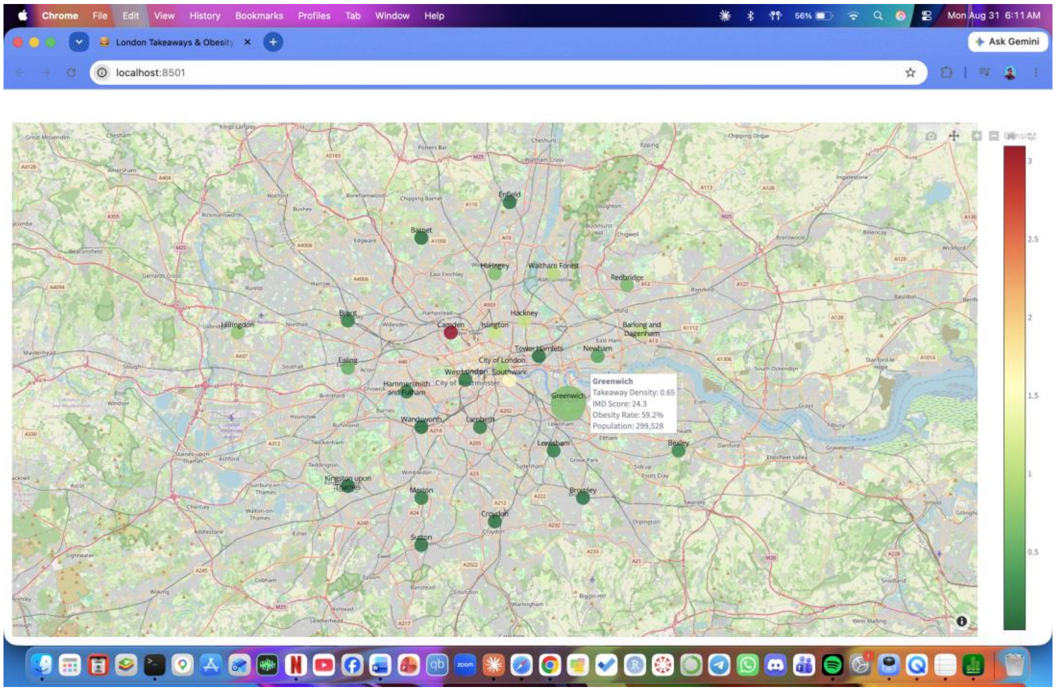

Figure 13: Dashboard Map View Extended

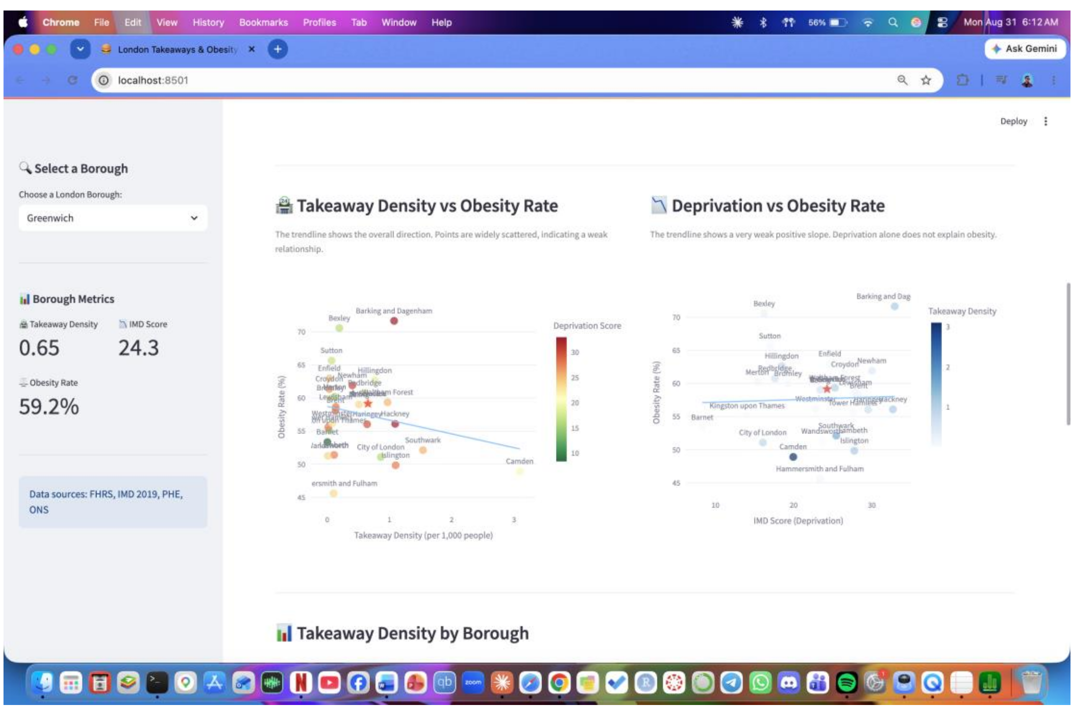

Figure 14: Dashboard Scatter Plot Trendline

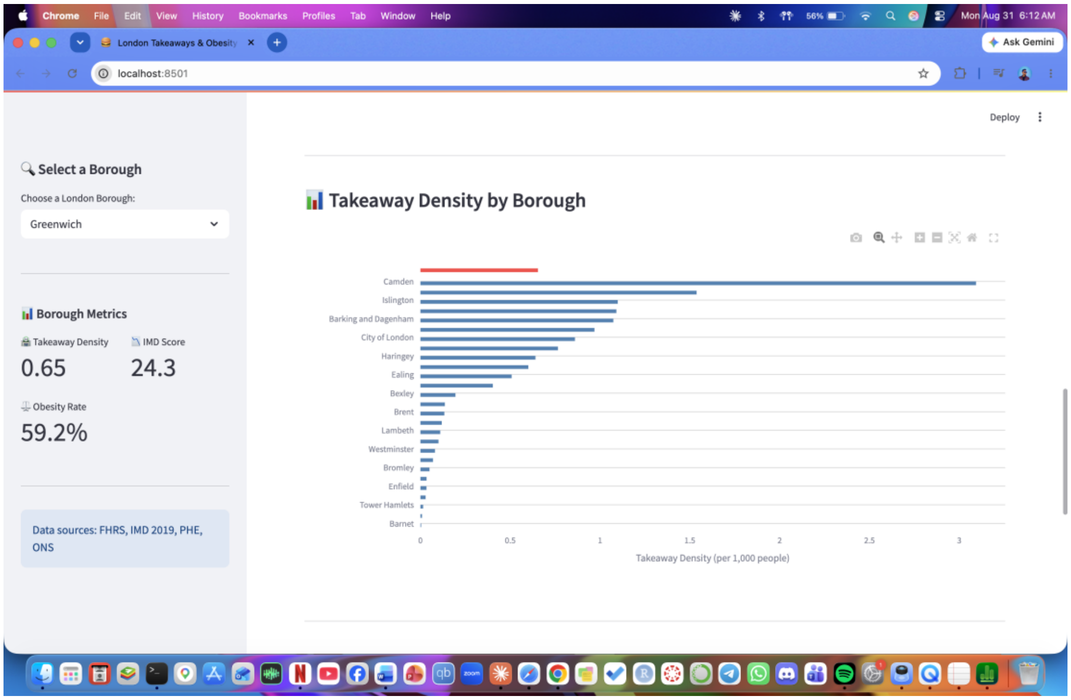

Figure 15: Dashboard Bar Chart

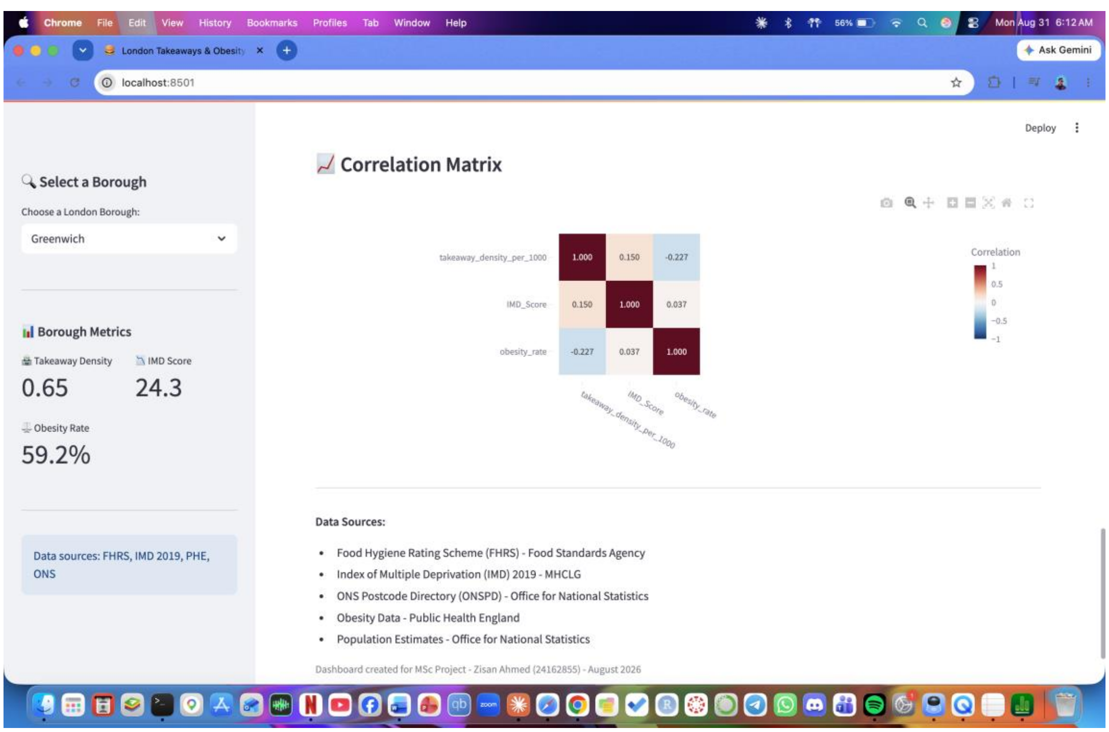

Figure 16: Dashboard Correlation Matrix

Key Dashboard Features:

Feature	Description
Borough Selector	Dropdown to select any of the 28 London boroughs
Key Metrics	Takeaway density, IMD score, obesity rate for selected borough
Interactive Map	Map showing takeaway density by borough with hover details
Scatter Plots	Three scatter plots with trendlines
Bar Chart	Takeaway density by borough with selected borough highlighted
Correlation Matrix	Heatmap showing correlations between variables

6. EVALUATION AND CONCLUSION

6.1 Summary of Findings

This investigation looked into the link between takeout density and obesity rates in London, specifically if deprivation explains the relationship. The main results are:

- There was no significant association between takeout density and obesity. Model A (takeaway density alone) explained only 5.1% of the variation in obesity rates, with a non-significant coefficient ($p = 0.246$).
- Including deprivation did not enhance the model. Model B (takeaway density plus deprivation) explained just 5.7% of the variance, and neither variable was significant ($p = 0.238$ for takeaways and $p = 0.715$ for deprivation).
- Both takeout and obesity have grown dramatically over time. The number of takeaways in London grew by 260.89 each year ($p = 0.003$), whereas obesity rates rose by 0.34 percentage points per year ($p = 0.002$).
- All regression assumptions were met. The models are statistically sound, even if the findings are not significant.

6.2 Evaluation of Research Question

The research question was:

To what extent does the density of takeaway food outlets in London's neighbourhoods predict obesity prevalence, after controlling for area-level deprivation?

Based on the results, the answer is: Takeaway density does not significantly predict obesity prevalence in London boroughs, even when deprivation is controlled for.

However, this does not necessarily mean there is no relationship. Several factors may explain the lack of significant findings:

- The sample size is small, with only 28 borough, which limits statistical power. This makes it tougher to discover a substantial association, even if one exists.
- Time mismatch: The takeout data is up to 2022, but the obesity data is up to 2025. This three-year delay might have hidden the relationship.
- Borough-level aggregation: The impact of food settings on obesity may be greatest at the neighborhood level. Aggregating to the boro level may have diluted this impact.
- Simple measure: This study just counted the number of takeout, not their quality, kind, or what people purchased. Not all takeaways are same.

6.3 Comparison with Literature

This study's findings contradict the meta-analysis of Pineda et al. (2024), which indicated a substantial positive relationship between fast-food density and obesity. However, the Pineda analysis comprised data from various geographic sizes, and the connection was typically larger at lower geographic levels.

The results are more consistent with previous research that revealed no connection when adjusting for socioeconomic characteristics (Cummins et al., 2014). This shows that the association between food settings and obesity is more nuanced than a straightforward density-outcome correlation.

The study further adds to the literature by emphasizing the significance of data alignment and geographical size. The temporal mismatch between datasets, as well as the borough-level aggregation, may have concealed a neighbourhood-level link.

6.5 Project Management

The project followed a structured approach, breaking the work into clear phases:

Phase	Tasks	Completed
1	Research and DPP	✅ May 2026
2	Data loading and cleaning	✅ June 2026
3	Data integration	✅ July 2026
4	Feature engineering and EDA	✅ July 2026
5	Regression modelling	✅ August 2026
6	Dashboard development	✅ August 2026
7	Final report writing	✅ September 2026

The project management strategy worked well, with frequent meetings with my supervisor to monitor progress and handle difficulties. The primary departure from the initial concept was the inclusion of the trend analysis, which was introduced in response to supervisor criticism.

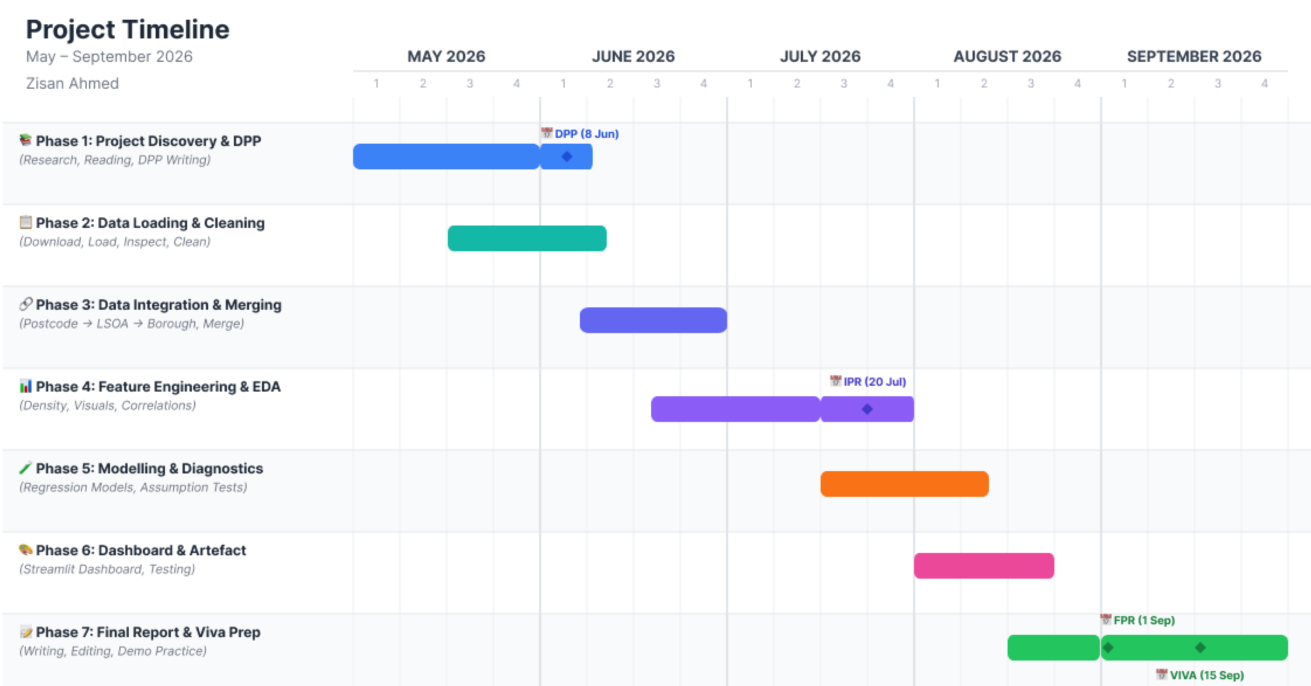

Figure 17: Grant Chart

Time management: The project was finished on time, however several activities took longer than projected (especially data integration and dashboard deployment). Allowing extra time for these chores was crucial.

6.6 Future Work

According on the findings and limitations of this study, future research could:

- Use more recent takeaway data: The FHRS data used in this analysis extends until 2022. More current data would lessen the temporal gap with obesity data.
- Analyze at the LSOA level: Borough-level grouping may obscure neighbourhood-level impacts. Future research might make use of LSOA-level obesity data if it is accessible.
- Differentiate between takeout types: Not all takeaways are the same. Differentiating by cuisine style, portion size, or nutritional composition may show more complex correlations.
- Include other food environment factors: Supermarkets, grocery shops, and access to fresh produce all have an impact on the food environment.
- Longitudinal analysis: Tracking changes over time may aid in determining causal linkages.
- Qualitative research: Understanding why consumers choose to dine at takeout restaurants and how the food environment impacts behavior should supplement quantitative data.

6.6.1 Potential Policy Implications

Although the study's conclusions are not statistically significant, they do have consequences for public health policy. If the link between takeaways and obesity is lower than previously assumed, then regulations limiting takeout outlets may have no effect on obesity trends. Policies that target the underlying causes of deprivation, such as income disparity, housing quality, and educational access, may prove more successful.

However, given the limitations of this investigation, these conclusions should be regarded with care. The limited sample size, temporal mismatch, and borough-level aggregation might have obscured a link that existed at the neighborhood level. Future study with more granular and recent data would be required to reach definite policy findings.

6.7 Conclusion

This investigation looked at the association between takeout density and obesity rates in London, using deprivation as a control variable. The investigation included five UK government datasets and used regression modeling and an interactive dashboard to investigate the correlations.

The study found that neither takeout density nor deprivation substantially predicted obesity rates in London boroughs. The findings contradict the general research, which has established favorable links between fast-food density and obesity. However, the findings are consistent with previous research that has demonstrated that the connection is weakened by socioeconomic characteristics and geographical size.

The study has significant drawbacks, including a limited sample size, timing differences between datasets, and borough-level aggregation. Despite these constraints, the study exhibits a thorough data science technique and adds to the literature on food environments and public health.

The interactive dashboard makes the findings more accessible to non-technical audiences and acts as a practical artifact highlighting the importance of data visualization in presenting research results. The dashboard is active and may be accessed online using the URL supplied in this report.

In conclusion, while the hypotheses were not confirmed, the findings show the complexities of the association between food settings and obesity, as well as the need of addressing data alignment, geographic scale, and socioeconomic characteristics in future studies. The project has given me excellent data science knowledge, such as data integration, statistical modeling, and dashboard construction, which will benefit me in my future job.

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APPENDICES

Appendix A: Final Dataset Structure

LocalAuthorityCode	takeaway_count	IMD_Score	BoroughName	population	takeaway_density_per_1000	obesity_rate
E09000001	13	16.07125	City of London	15111	0.8603004433856130	51.07349
E09000002	250	32.917840000000000	Barking and Dagenham	232747	1.074127700894100	71.65536
E09000003	1	8.319	Barnet	405050	0.002468831008517470	53.34249
E09000004	50	16.2332	Bexley	256434	0.1949819446719230	70.54951
E09000005	47	28.31002631578950	Brent	352976	0.13315352885182000	58.15166

E09000006	17	19.284785714285700	Bromley	335319	0.05069799206129090	59.9997
E09000007	671	19.947173469387800	Camden	216943	3.092978339932610	48.90008
E09000008	14	26.56346153846150	Croydon	409342	0.0342012302671116	61.35185
E09000009	196	23.399448275862100	Ealing	385985	0.5077917535655530	59.03182
E09000010	11	24.638454545454500	Enfield	327434	0.03359455646023320	62.97327
E09000011	196	24.26660810810810	Greenwich	299528	0.6543628642397370	59.1521
E09000012	291	32.68148837209300	Hackney	266758	1.0908763748416200	56.09263
E09000013	19	23.3695	Hammersmith and Fulham	188687	0.10069586140009600	45.58075
E09000014	169	29.516791666666700	Haringey	263850	0.6405154443812770	56.03712
E09000017	252	18.478875000000000	Hillingdon	329185	0.7655269833072590	62.65492
E09000019	245	27.750649350649400	Islington	223024	1.0985364803787900	49.83933
E09000021	5	14.048000000000000	Kingston upon Thames	172692	0.02895328098580130	55.1374
E09000022	35	27.572633333333300	Lambeth	316920	0.11043796541714000	51.40355
E09000023	41	27.891125	Lewisham	301255	0.1360973261854580	58.64016
E09000024	26	15.344722222222200	Merton	218539	0.11897189975244700	60.17755
E09000025	151	29.99673529411770	Newham	374523	0.4031795110046650	61.9076
E09000026	193	17.60870588235290	Redbridge	321231	0.6008137446261410	60.76359
E09000028	484	25.43720720720720	Southwark	314786	1.537552495981400	52.12476
E09000029	15	16.970600000000000	Sutton	214525	0.06992192052208370	65.64052
E09000030	5	27.542	Tower Hamlets	331886	0.015065414027708300	55.57481
E09000031	271	25.266463917525800	Waltham Forest	279737	0.9687670919470790	59.35304
E09000032	3	23.634	Wandsworth	337655	0.008884808458337650	51.26103
E09000033	17	22.83866666666670	Westminster	209996	0.08095392293186540	56.17029

Appendix B: Regression Outputs

Model A Output:

```
==============================================================
MODEL A: BIVARIATE REGRESSION
Takeaway Density → Obesity Rate
==============================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:           obesity_rate   R-squared:                       0.051
Model:                            OLS   Adj. R-squared:                  0.015
Method:                 Least Squares   F-statistic:                     1.410
Date:                Sun, 30 Aug 2026   Prob (F-statistic):              0.246
Time:                        16:34:47   Log-Likelihood:                -89.254
No. Observations:                  28   AIC:                             182.5
Df Residuals:                      26   BIC:                             185.2
Df Model:                           1
Covariance Type:            nonrobust
=============================================================================================
                               coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------------
const                       58.7325      1.461     40.197      0.000      55.729      61.736
takeaway_density_per_1000   -2.0792      1.751     -1.187      0.246      -5.679       1.520
==============================================================================
Omnibus:                        2.374   Durbin-Watson:                   2.497
Prob(Omnibus):                  0.305   Jarque-Bera (JB):                1.131
Skew:                           0.408   Prob(JB):                        0.568
Kurtosis:                       3.551   Cond. No.                         2.11
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

==============================================================
MODEL A - KEY METRICS
==============================================================
R-squared: 0.0514
Adjusted R-squared: 0.0150
F-statistic: 1.4098
F-statistic p-value: 0.2458
Number of observations: 28
AIC: 182.51
BIC: 185.17

--------------------------------------------------------------
COEFFICIENTS:
--------------------------------------------------------------
Intercept (β0): 58.7325
Takeaway Density (β1): -2.0792
p-value for β1: 0.2458
⚠️ The coefficient is NOT statistically significant (p >= 0.05).
```

Model B Output:

```
============================================================
MODEL B: MULTIVARIATE REGRESSION
Takeaway Density + Deprivation → Obesity Rate
============================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:           obesity_rate   R-squared:                       0.057
Model:                            OLS   Adj. R-squared:                 -0.019
Method:                 Least Squares   F-statistic:                    0.7499
Date:                Sun, 30 Aug 2026   Prob (F-statistic):              0.483
Time:                        16:34:47   Log-Likelihood:                -89.177
No. Observations:                  28   AIC:                             184.4
Df Residuals:                      25   BIC:                             188.4
Df Model:                           2
Covariance Type:            nonrobust
=============================================================================================
                               coef    std err          t      P>|t|      [0.025      0.975]
---------------------------------------------------------------------------------------------
const                       57.0836      4.701     12.142      0.000      47.401      66.766
takeaway_density_per_1000   -2.1790      1.801     -1.210      0.238      -5.889       1.531
IMD_Score                    0.0737      0.199      0.370      0.715      -0.337       0.484
==============================================================================
Omnibus:                        2.106   Durbin-Watson:                   2.442
Prob(Omnibus):                  0.349   Jarque-Bera (JB):                0.961
Skew:                           0.385   Prob(JB):                        0.618
Kurtosis:                       3.480   Cond. No.                         96.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

============================================================
MODEL B - KEY METRICS
============================================================
R-squared: 0.0566
Adjusted R-squared: -0.0189
F-statistic: 0.7499
F-statistic p-value: 0.4828
Number of observations: 28
AIC: 184.35
BIC: 188.35

------------------------------------------------------------
COEFFICIENTS:
------------------------------------------------------------
Intercept (β0): 57.0836
Takeaway Density (β1): -2.1790
p-value for β1: 0.2377
IMD Score (β2): 0.0737
p-value for β2: 0.7147
⚠️ Takeaway density coefficient is NOT statistically significant (p >= 0.05).
⚠️ IMD Score coefficient is NOT statistically significant (p >= 0.05).
```

Appendix C: Figures and Screenshots

Figure 1: Correlation Matrix

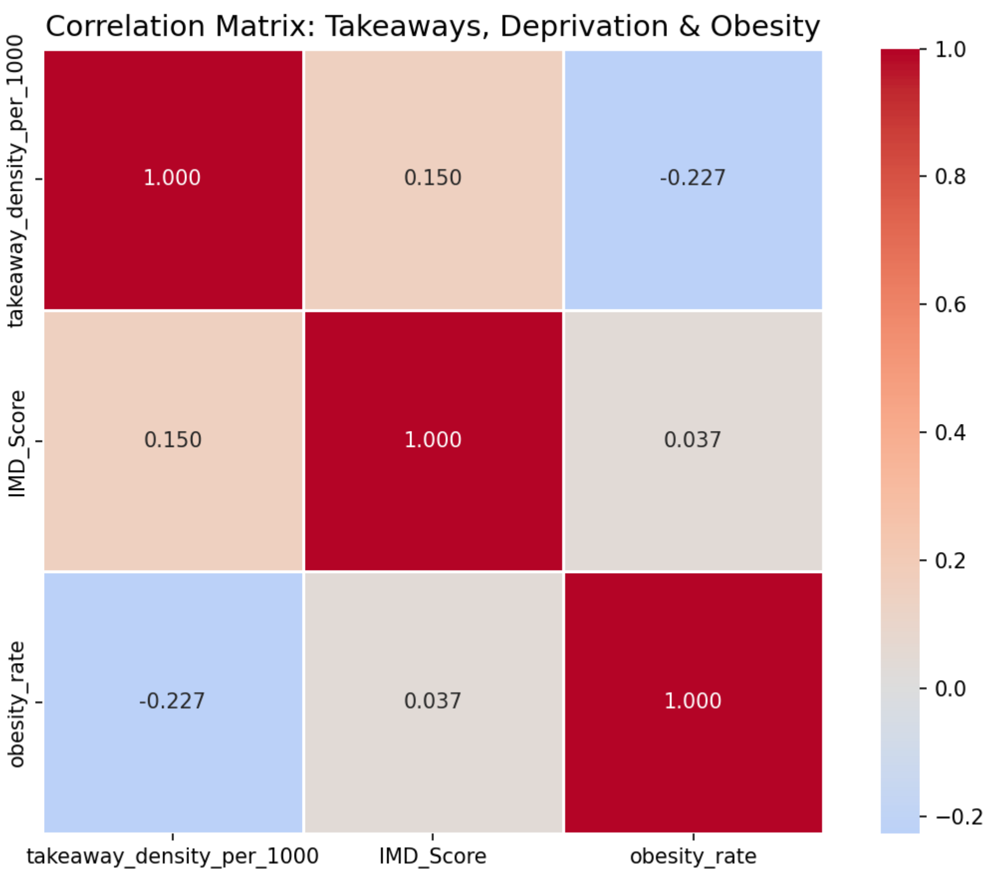

Figure 2: Scatterplots

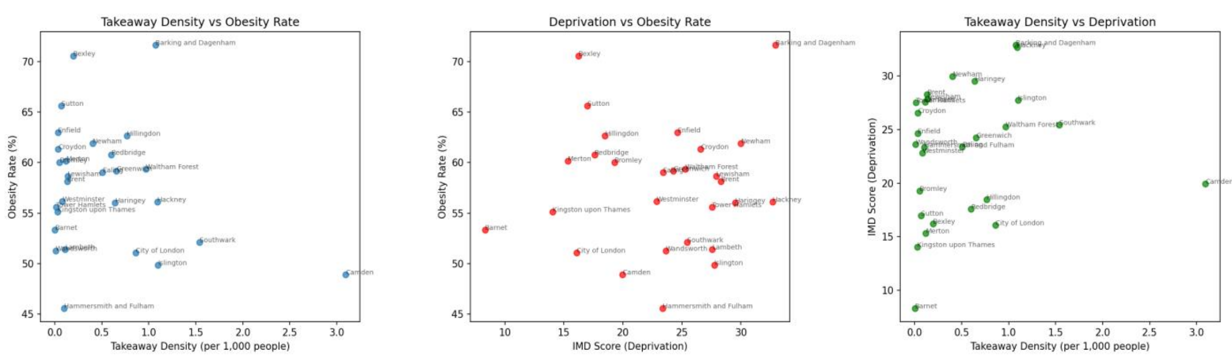

Figure 3: Scatter Plots with Trendlines.

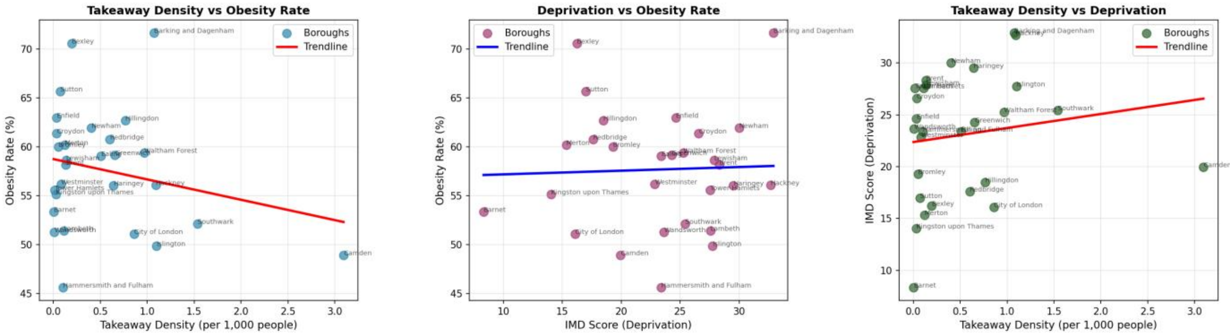

Figure 4: Takeaway density by borough.

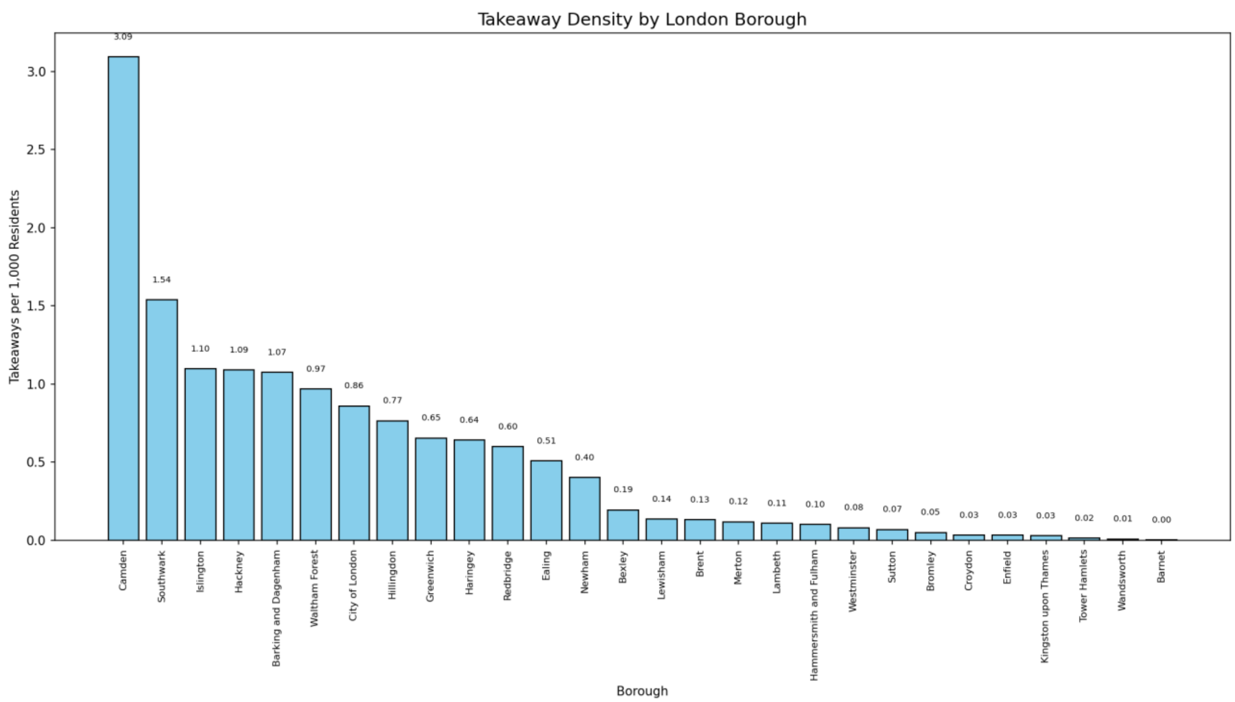

Figure 5: The number of takeaways in London over time.

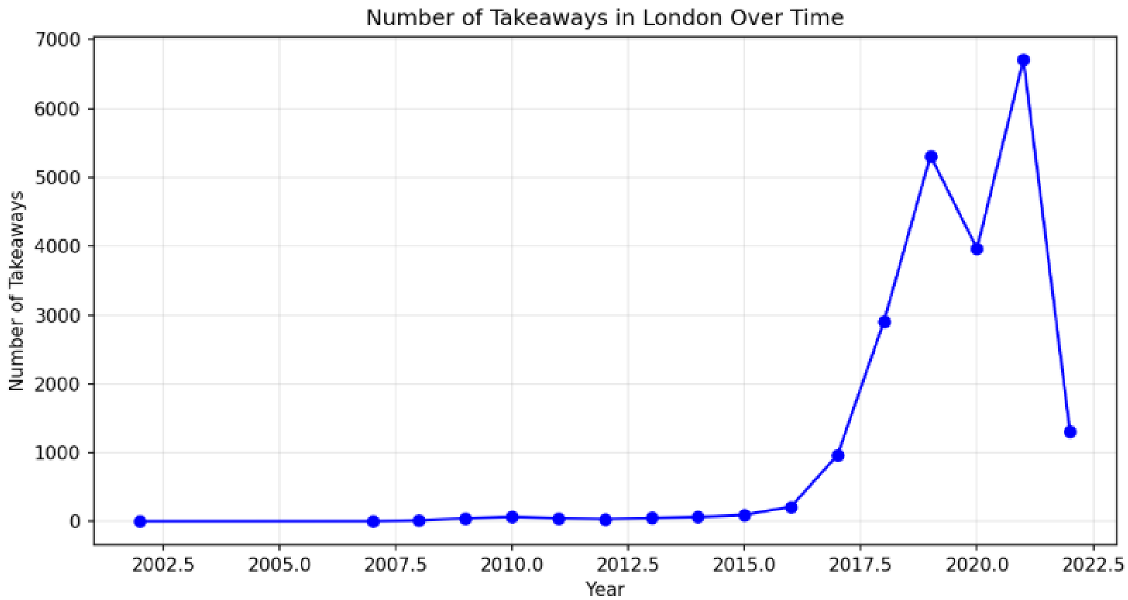

Figure 6: The average obesity rate in London throughout time.

Figure 7: Linearity Check.

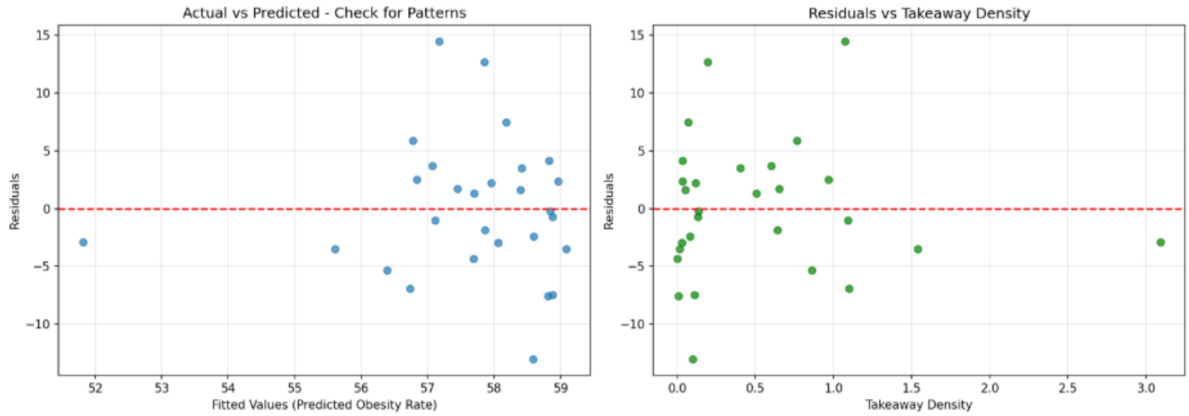

Figure 8: Normality Check.

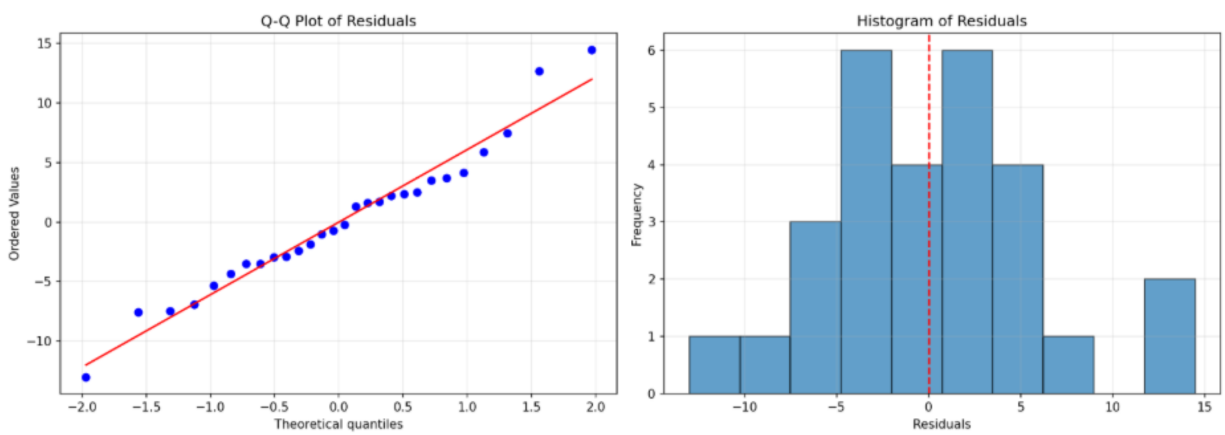

Figure 9: Homoscedasticity Check.

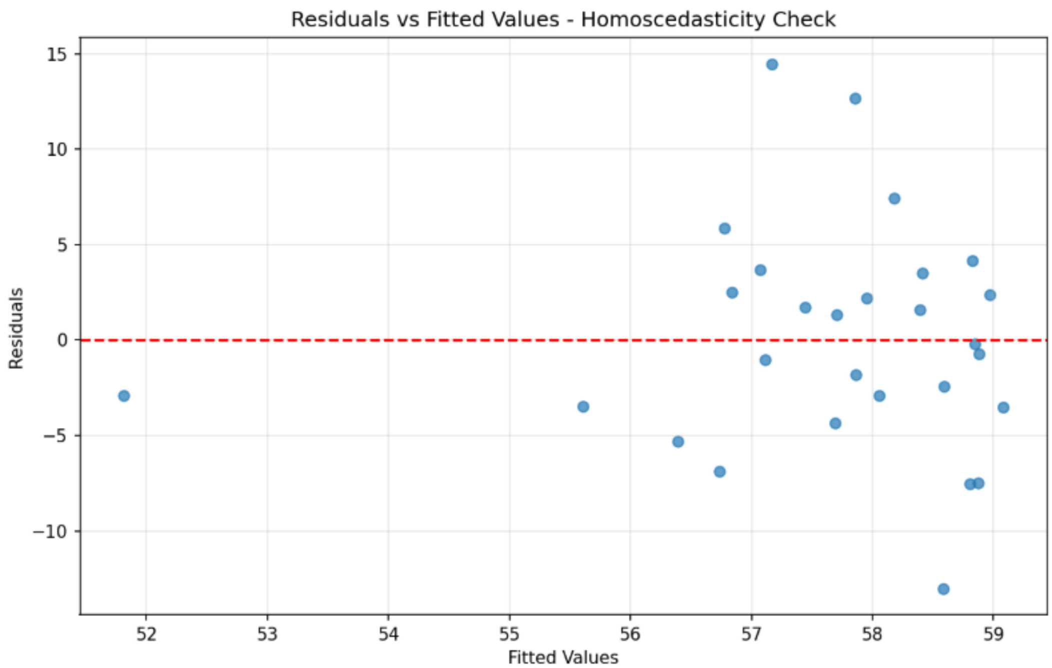

Figure 10: Independence Check.

Figure 11: Dashboard Main View

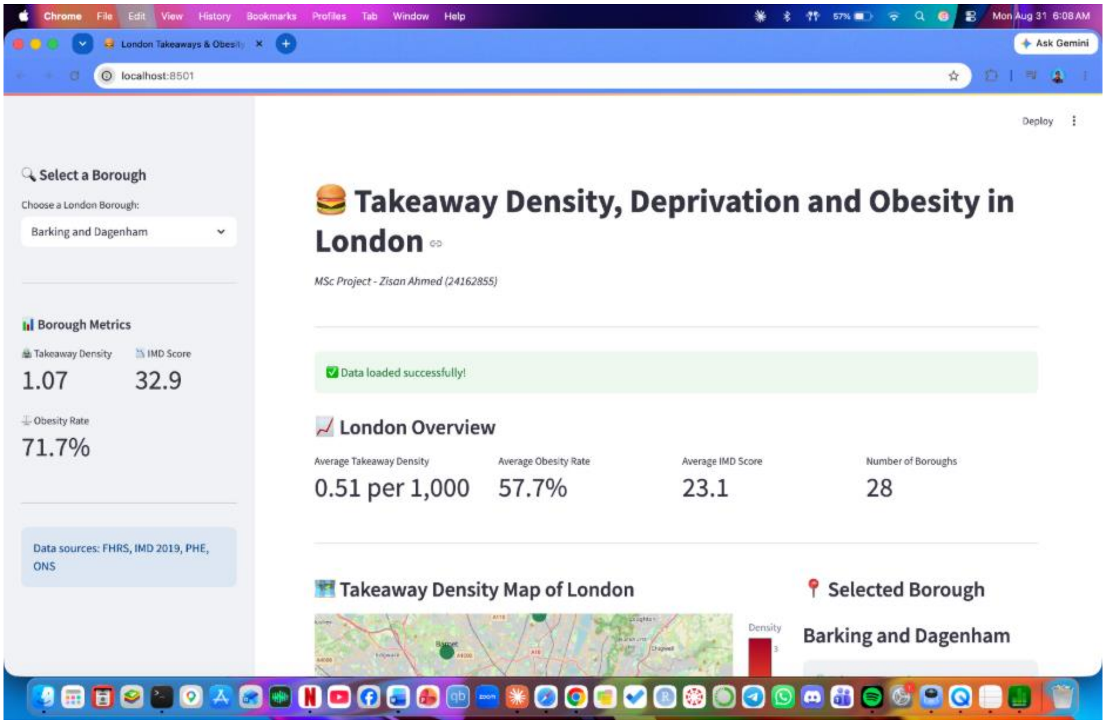

Figure 12: Dashboard Map View

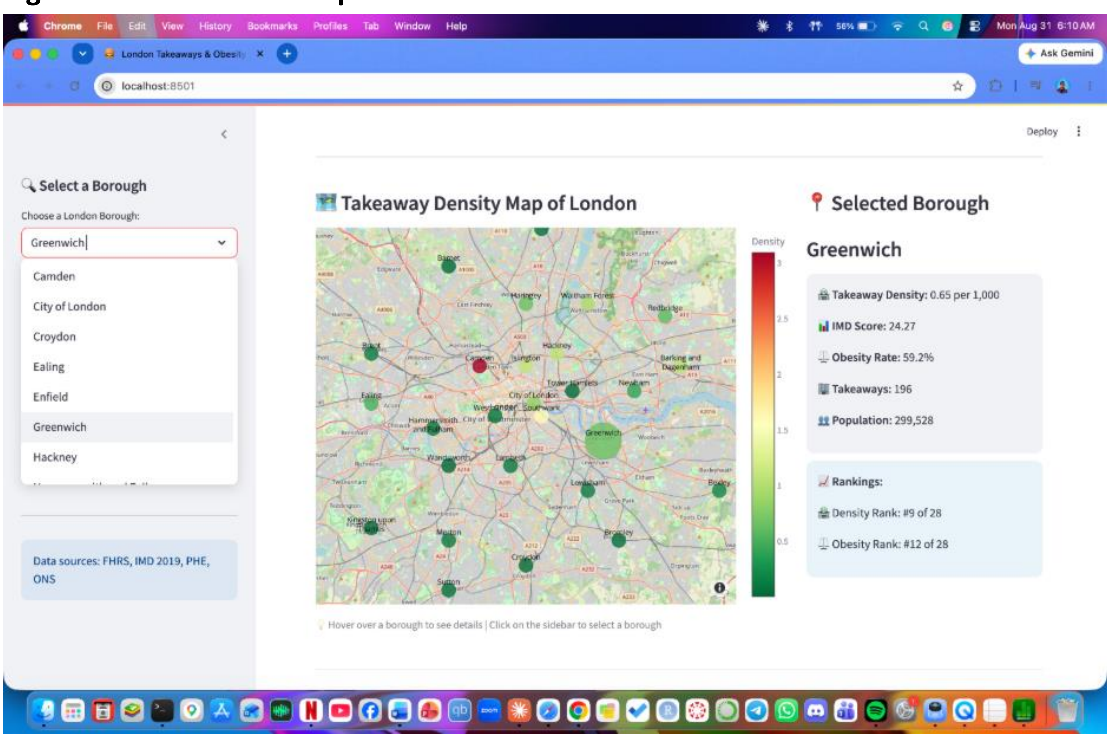

Figure 13: Dashboard Map View Extended

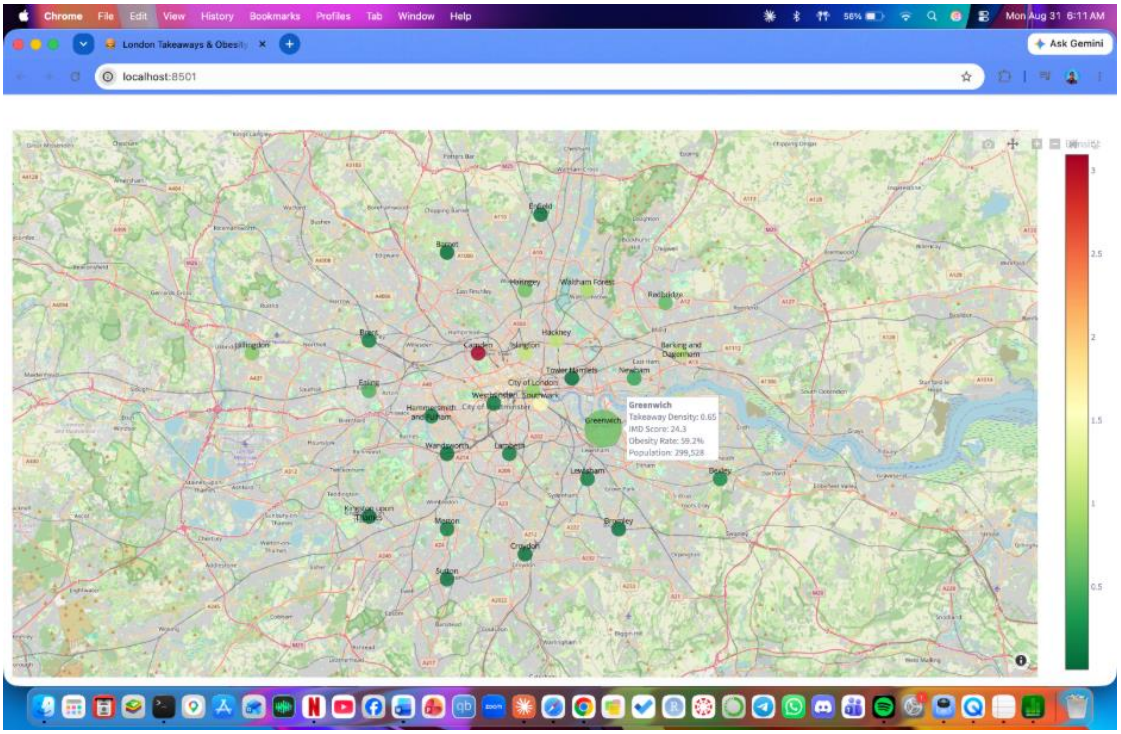

Figure 14: Dashboard Scatter Plot Trendline

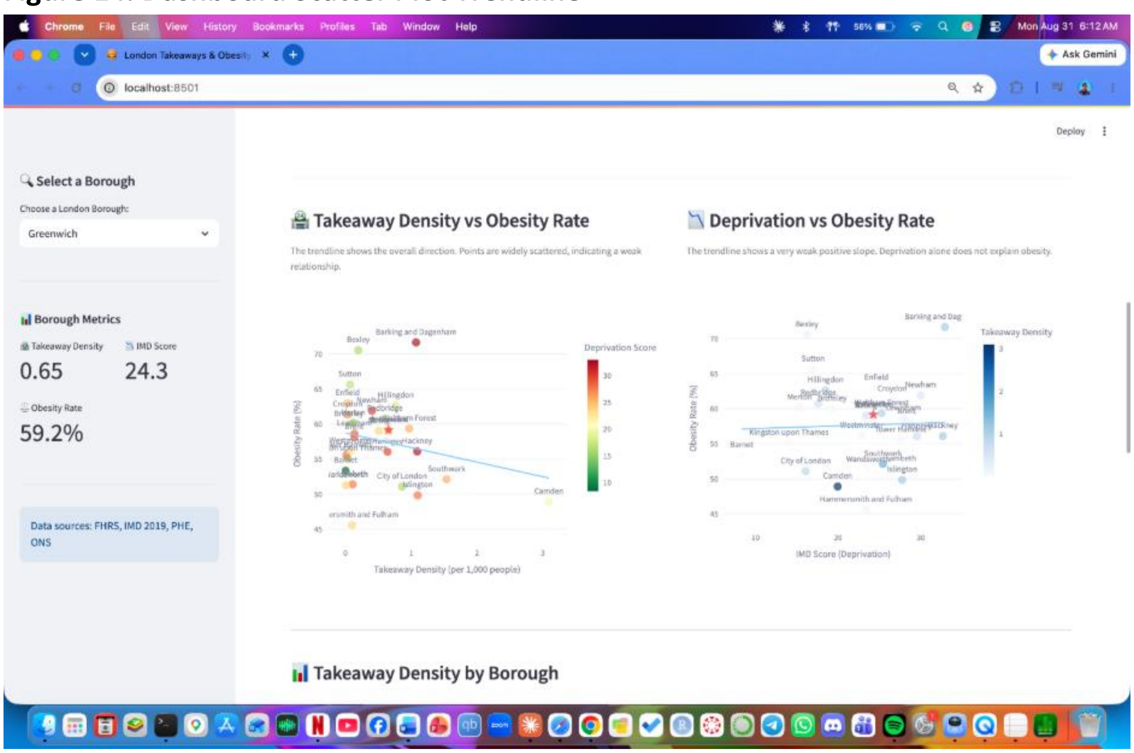

Figure 15: Dashboard Bar Chart

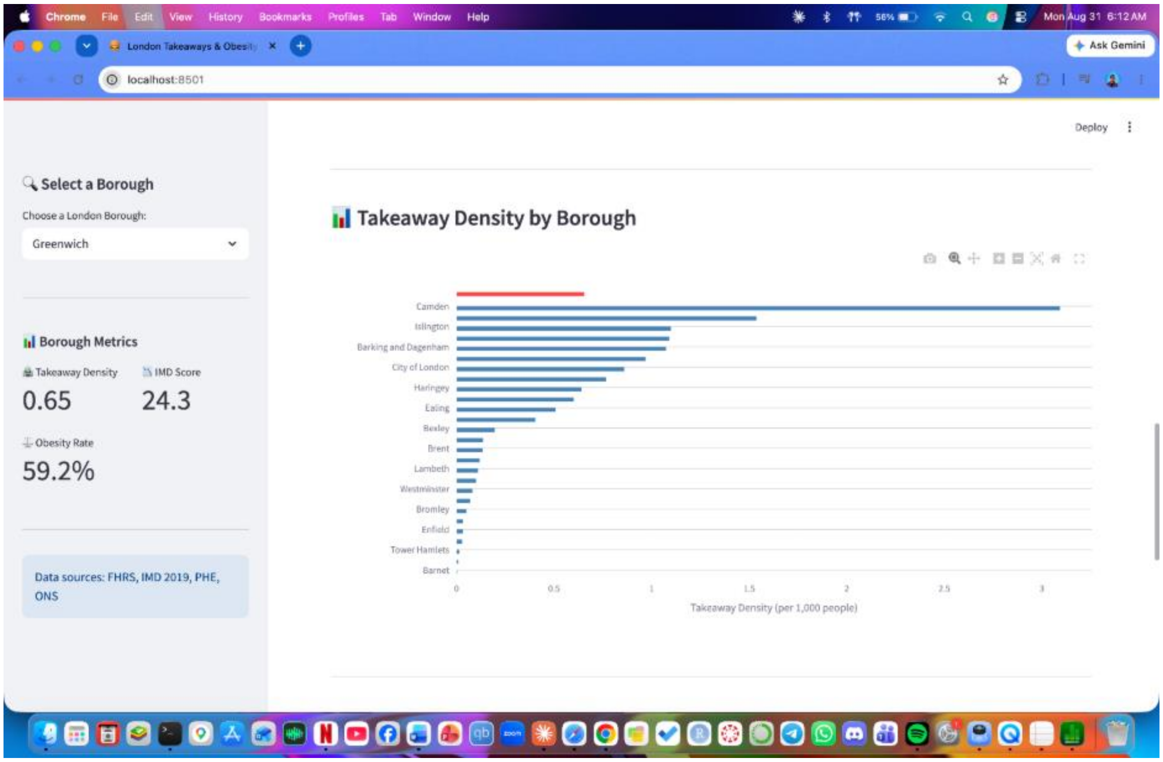

Figure 16: Dashboard Correlation Matrix

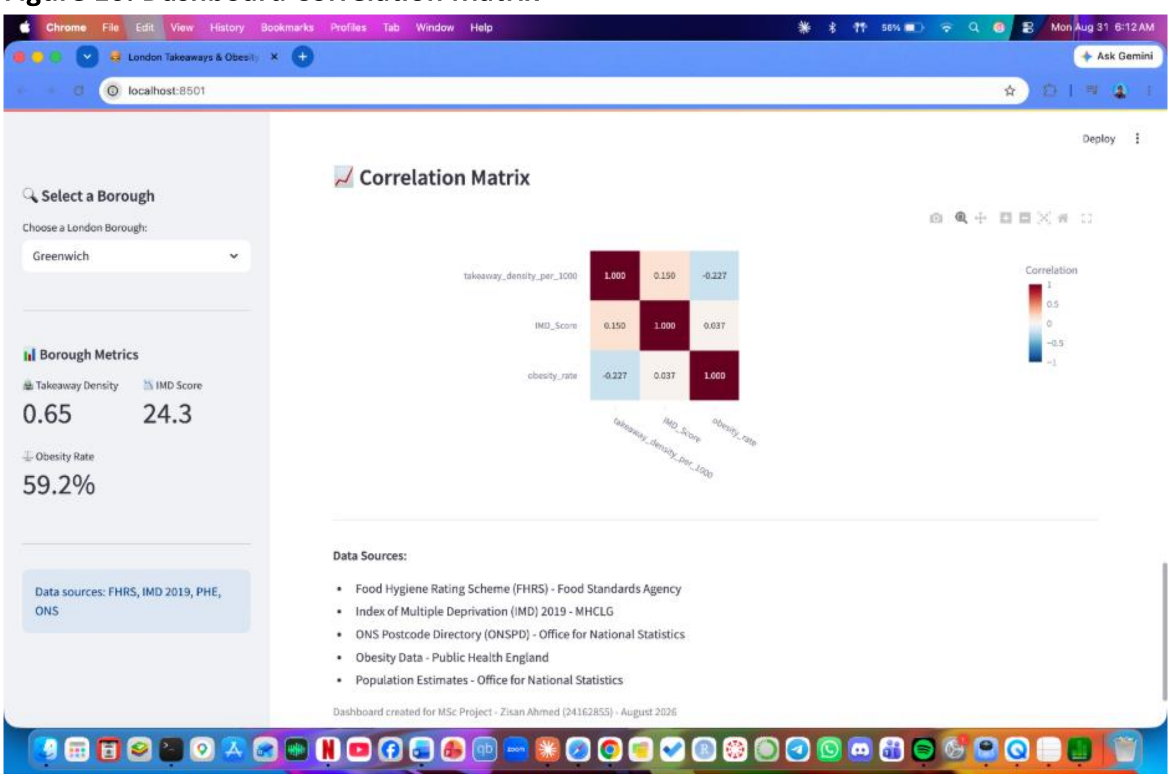

Figure 16: Grant Chart